Artifact Types Catalog

ICUHeart - Artifact Detection Project

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Abstract

This report investigates the influence of each of the 18 artifact types on the three ICU physiological signals: ECG, BP and CVP.

For each artifact type we provide three tables: 'Artifact Summary' lists IDs for patient, artifact, task and task step. This provides a link to be used with the `Artifact_Plots_Catalog.pdf` file; 'Artifact Notes' lists all notes recorded in the `Artifact_Plots_Catalog.pdf` file; 'Notes Processing Summary' lists the assessed effect on the physiologic and accelerometer signals as well as a note whether or not the artifact has been missed or appears pre or post the annotation carried out at the bed side.

For each artifact type a 'Signal Summary Plot' shows the effect on each signal type.

1 Introduction

This document provides a detailed analysis of the effects of 18 artifact types on the ICU physiologic signals ECG, BP and CVP.

This document should be read in conjunction with the document `Artifact_Plots_Catalog.pdf` (Piper and Donald (2023)).

The data was collected for a pilot study run by the 'ICU Heart' project (Docherty (2023)). This is described in Sandall (2022). This work identified 152 annotations across seven patients. A further three patients were added to the data set in late 2021 giving a total of 235 annotated events. Dr Annemarie Docherty assessed these events and identified 151 as artifactual with 84 as having some physiological basis. This document provides an analysis of these 151 events. One event, a blood sample from patient 7 was lost due to corrupted data.

Demographic information is provided for the first seven patients from Table 2 in Katerina Sandall's report (*reproduced in Table 1*).

Table 1: Demographics for patients 1 to 7

Patient	Age	Gender	History
1	76	Male	OOHCA
2	74	Male	Polytrauma, RTC, type B aortic dissection
3	65	Male	SAH, stroke
4	55	Female	Status epilepticus, autoimmune encephalitis
5	47	Male	Guillain-Barré syndrome
6	69	Male	OOHCA, previous bypass surgery
7	54	Female	SAH, breast cancer

- Abbreviations: OOHCA, Out of Hospital Cardiac Arrest; RTC, Road Traffic Collision; SAH, subarachnoid haemorrhage.

2 All Artifact Types

2.1 Summary Numbers

Table 2: Artifact Type Total Numbers

Artifact Type	N	Num Patients
Washing body	22	6
Blood_sample	20	9
Chest_abdo_limb_exam	14	7
Bed_movement	11	7
Turn	11	6
Adjusting_lines	11	5
Gown_change	11	4
IV_medication	11	4
ECG_lead	9	3
Pupils	7	2
Compression_sock_adjust	6	1
Ng_feed	4	2
Temperature	3	3
Subglottic_drainage	3	2
Patient_moving	2	2
Reposition_limb	2	2
BP_cuff	2	1
Transducer_calibration_alteration	2	1

The above table is arranged by N, the number of artifact examples, and then number of patients. Where there is a tie, the artifact type is arranged alphabetically.

Table 3: Artifact Types Total Numbers By Patient ID

Patient ID	N
8	39
9	35
7	21
6	15
2	9
4	8
3	7
1	6
5	6
10	5

Note: In Table 3, Patient 7 has 21 artifacts recorded in the file `EVENT_DURATIONS2.csv`. However the data for the last artifact, a blood sample, seems to be corrupted therefore we only have details on 20 artifacts.

2.2 Artifact Distribution By Patient

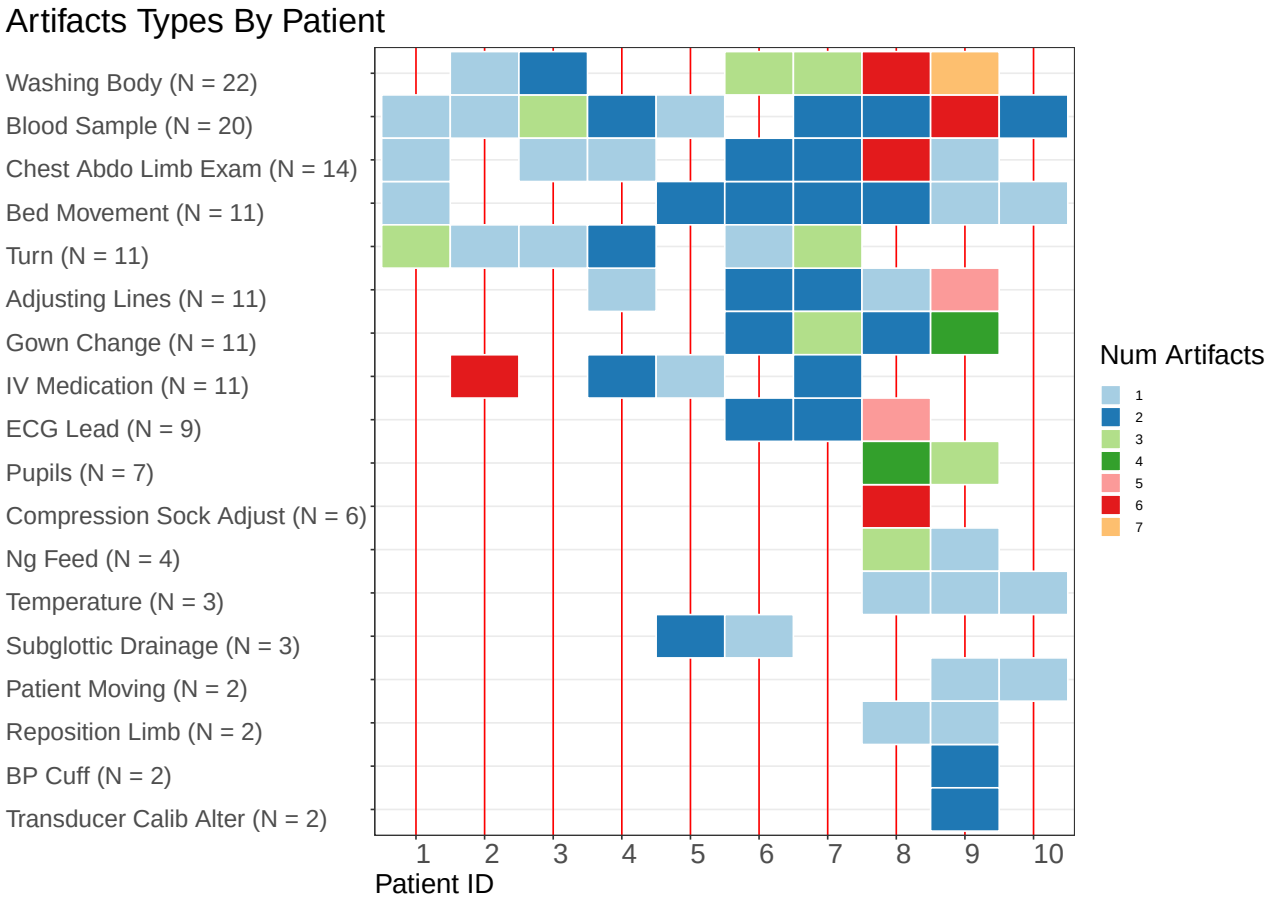

Figure 1: HeatMap: Artifact Types By Patient

2.3 Signal Effect Summaries

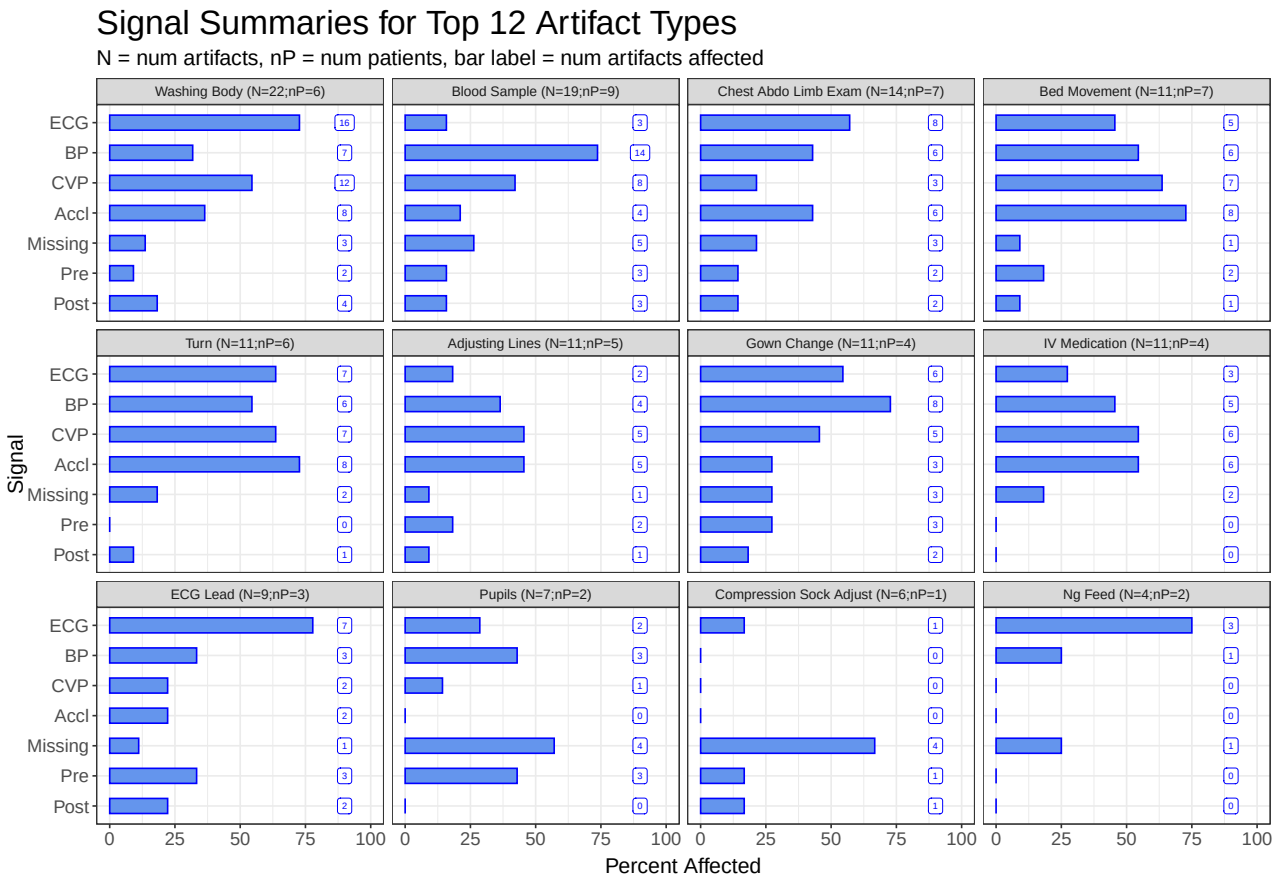

Figure 2: Signal Summaries for Top 12 Artifact Types

In Figure 2 we show the effect of a given artifact on the three physiological signals ECG, BP and CVP. The 'Accl' signal indicates whether there was any fluctuation on any of the accelerometer planes. We have observed that there are annotation periods where there is no artifact present this is the 'Missing' signal. There are also a number of cases where there is clearly artifactual data but it is before, the 'Pre' signal, or after, the 'Post' signal, the annotation period.

2.4 Table and Plot Information

In the sections that follow each artifact type has three tables.

- Artifact Summary: This lists the patient ID as well as the artifact number. It also shows which task and task step has been identified. This provides a link to be used with the `Artifact_Plots_Catalog.pdf` file.

- Artifact Notes: This lists all the notes recorded as we examined each artifact for the `Artifact_Plots_Catalog.pdf` file. Each note is provided with a cross reference which links it with the summary table and the plots catalog. The `EPN` column is the (E)xtra (P)lot (N)umber if present.
- Notes Processing Summary: This lists the assessed effect on the physiologic and accelerometer signals as well as a note whether or not the artifact has been missed or appears pre or post the annotation carried out at the bed side. A one indicates some effect, a zero indicates no effect.

In addition there is one 'Signal Summary Plot' which visualises the effect on each signal type.

3 Washing Body (N = 22, Number of Patients = 6)

3.1 Summary

Table 4: Artifact: Washing Body Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Washing body	2	7	5	1	7 mins 15.6 secs
	3	3	3	1	31.1 secs
	3	6	6	1	1 min 2.2 secs
	6	6	3	3	1 min 56.5 secs
	6	7	3	4	1 min 41.2 secs
	6	8	3	5	10 secs
	7	10	6	3	1 min 40 secs
	7	11	6	4	55 secs
	7	14	6	7	45 secs
	8	19	10	1	4 secs
	8	24	10	6	49.2 secs
	8	25	10	7	53.2 secs
	8	27	10	9	38.2 secs
	8	30	10	12	24.1 secs
	8	32	10	14	4 secs
	9	14	8	1	16.1 secs
	9	17	8	4	28.1 secs
	9	18	8	5	37.1 secs
	9	19	8	6	51.2 secs
	9	20	8	7	38.2 secs
	9	29	9	7	59.2 secs
	9	30	9	8	23.1 secs

3.2 Notes

Table 5: Artifact: Washing Body Notes }

PID	ArtID	Task	Step	EPN	Note Text
2	7	5	1	-	1. Clear changes in CVP that maps to the BP signal. Two bursts. 2. ECG lost during the body washing event suggesting it is chest washing in progress.
3	3	3	1	-	1. Very little effect here, except near the end of the event window with some extra beats in ECG/BP. See the Zoomed in plot below.

Table 5: *(Artifact: Washing Body ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
3	3	3	1	1	1. See the extra ECG/BP events between green and red cursor. The fact that both BP and ECG were effected suggests that the event must have triggered a minor physiological response?
3	6	6	1	-	1. One can see another example of washing body events being picked up in the CVP signal with some short ”Chirp” like noise in some of the accelerometer planes. 2. Note the phase shift between the accelerometer and CVP.
6	6	3	3	-	1. First of 3 washing body events (are they linked?). 2. Some clear noise on CVP and ECG together with Accelerometer noise.
6	7	3	4	-	1. Similar type responses in the signals as with previous body wash event.
6	8	3	5	-	1. NOT typical body-wash event. No obvious effects on any signals. Did Kat miss the event?
7	10	6	3	-	1. Washing body effects on BP and ECG lines at two different times during the event. 2. x,y accelerometer changes to be expected with washing the chest? 3. See the zoomed in plot below to look at how the movement affects ecg/bp.
7	10	6	3	1	1. see the effects of the washing body movement on the ECG and BP signals between the red and green cursors.
7	11	6	4	-	1. More subtle changes in ECG/BP signals during washing body showing a slow sinusoidal fluctuation in baseline. Is this what happens when washing another non-chest part of the body? 2. Expected changes in x,y planes of accelerometer.
7	14	6	7	-	1. Washing body only effects ECG trace here. x,y plane movement seen too.
8	19	10	1	-	Obvious signals on the all the accelerometer channels. Oddly no effect on the physiological channels.
8	24	10	6	-	A relatively long annotation (almost a minute) which shows artifact patterns on the ECG and BP signals. The four cursors indicate possible regions for further investigation. The CVP signal has the pumped infusion pattern. The accelerometers do not show any unusual patterns.

Table 5: *(Artifact: Washing Body … continued)*

PID	ArtID	Task	Step	EPN	Note Text
8	25	10	7	-	Another relatively long annotation (almost a minute) which shows artifact patterns on the ECG and BP signals at the green cursor. The red cursor shows a possible artifact only on the BP signal. The blue cursor shows an artifact pattern which appears on the CVP signal and possibly the ECG signal. This is most likely PIIA. At the start of the annotation period the CVP signal is showing the pumped infusion pattern however it changes to a lower more steady signal at the yellow cursor albeit still with a pump spike.
8	27	10	9	-	ECG trace missing at the start of annotation. Need to add about 1.5 mins to the start to see the initial ECG disconnection.
8	30	10	12	-	Single CVP artifact starting about 3334984. Possible ECG artifacts starting about 3337097 and continuing past the annotation end time.
8	30	10	12	1	Zoomed into possible ECG artifacts starting about 3336967 and continuing past the annotation end time.
8	32	10	14	-	ECG trace seems very odd, possibly suggesting the ECG lead has been affected by the washing body procedure. ECG also odd pre and post annotation time. The BP trace shows artifactual effects. In particular note the reducing time between BP trace systolic peaks indicating an increasing heart rate. The CVP signal has two raised areas (the red ovals).
9	14	8	1	-	No obvious artifact. Task 8, start of a nine step sequence, ID 14 to 22.
9	17	8	4	-	No obvious artifact.
9	18	8	5	-	Possible artifact on CVP line, see green and red cursors. After a review this is not considered to be artifact linked to the washing body procedure. Zoomed in to first possible artifact on next plot @fig-patient-9-Artifact-18-EP-01.
9	18	8	5	1	Zoomed in to first possible artifact for Artifact 18. This plot shows the disturbance on CVP signal and a possible coincident disturbance on the BP signal. After a review this is not considered to be artifact linked to the washing body procedure.
9	19	8	6	-	Multiple artifacts on CVP and BP signals. In one case (green cursor), a disturbance across all three physiological signals. Interestingly no obvious disturbance on the accelerometer signals. See @fig-patient-9-Artifact-19-EP-01 for zoomed in plot around the green cursor.

Table 5: *(Artifact: Washing Body … continued)*

PID	ArtID	Task	Step	EPN	Note Text
9	19	8	6	1	This plot shows a disturbance across all three physiological signals. The ECG artifact seems to lag the BP and CVP signals.
9	20	8	7	-	After a review this is not considered to be artifact linked to the washing body procedure. Possible CVP artifact between the green and red cursors. There is also a hint of artifact in a raised BP peak. Again no obvious artifact on the accelerometer signals. Further examination in the next two plots @fig-patient-9-Artifact-20-EP-01 and @fig-patient-9-Artifact-20-EP-02.
9	20	8	7	1	After a review this is not considered to be artifact linked to the washing body procedure. This is a zoomed in plot around the green cursor from the previous plot @fig-patient-9-Artifact-20. Note that the CVP seems to rise and there is also an increase in the coincident BP peak. *In this next bit, remember I'm no medical doctor so this might be wildly over interpreting things! I also may not be correctly identifying the ECG segments.* If you look very closely you can see two odd looking ECG pulses as well with *rounded* sections from the 'J' point to next 'P' wave (ST interval + TP interval). These are examined in the next plot @fig-patient-9-Artifact-20-EP-02.
9	20	8	7	2	After a review this is not considered to be artifact linked to the washing body procedure. *Again, remember I'm no medical doctor so this might be way off!* In this ECG plot I have highlighted two areas as having *rounded* sections from the 'J' point to next 'P' wave (ST interval + TP interval). The ECG plot does not carry through the cursors but you can count seven 'R' peaks from (and including) the 0 point. If you look back at the first extra plot @fig-patient-9-Artifact-20-EP-01 and count seven peaks from the end of the red pre-shading you can track the correspondence. After this seventh peak, you see the first *rounded* section. It seems to me that in all the other similar sections, the signals slopes downwards or remain flat from the 'J' point. The same thing appears to happen after the ninth peak.
9	29	9	7	-	After a review this is not considered to be artifact linked to the washing body procedure. Possible artifact on BP signal between green and red cursors. Disturbance in the CVP signal between blue and yellow cursors. Nothing on ECG or the accelerometer signals.

Table 5: *(Artifact: Washing Body ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
9	30	9	8	-	Multiple artifact patterns on ECG and CVP signals. Nothing on the BP or accelerometer signals. This is the last step in Task 9.

3.3 Signal Summary

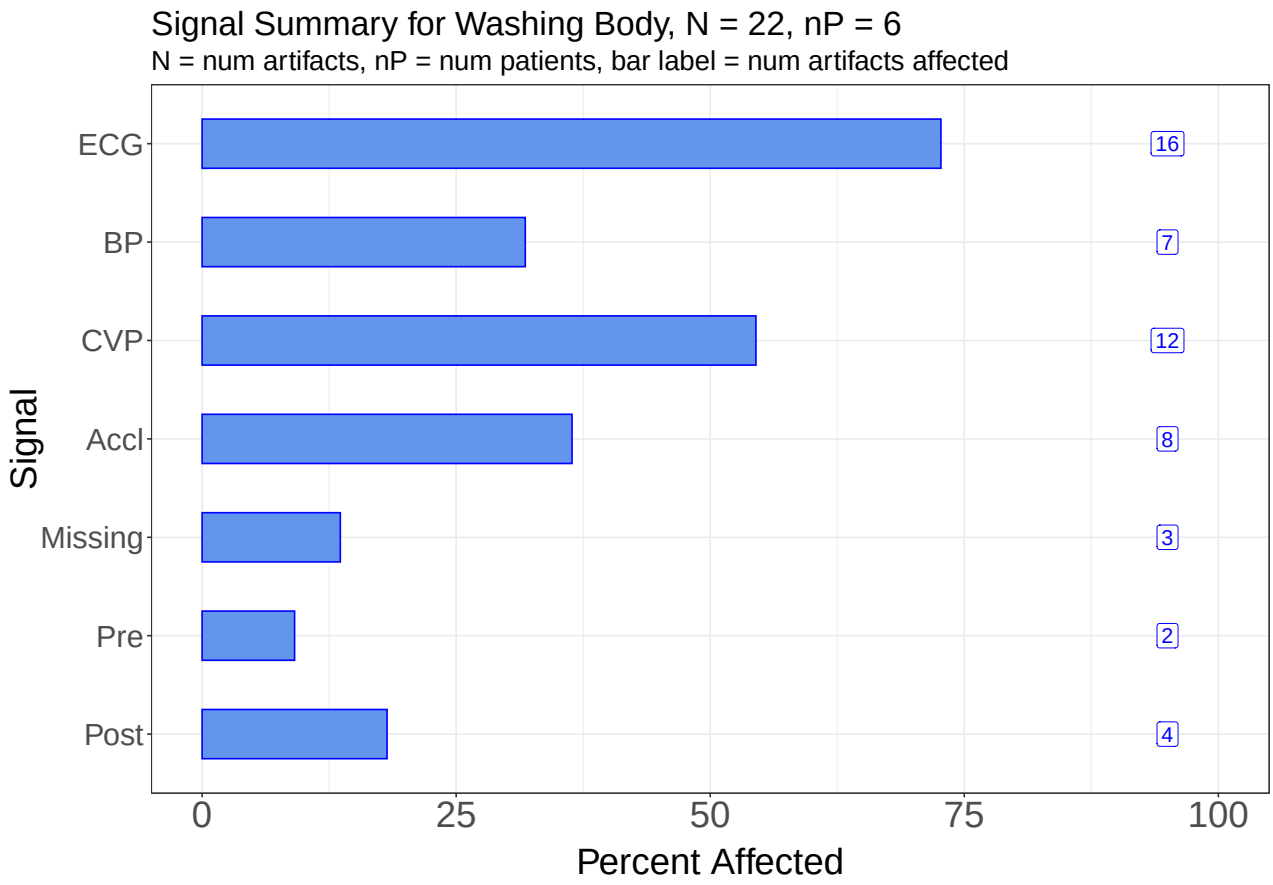

Figure 3: Artifact: Washing Body Notes: Signal Summary

3.4 Notes Processing Summary

Table 6: Artifact: Washing Body Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
2	7	5	1	1	1	1	1	0	0	0
3	3	3	1	1	1	0	0	0	0	1
3	6	6	1	1	1	1	1	0	0	0
6	6	3	3	1	0	1	1	0	0	0
6	7	3	4	1	1	1	1	0	0	0
6	8	3	5	0	0	0	0	1	0	0
7	10	6	3	1	1	0	1	0	0	0
7	11	6	4	1	0	0	1	0	0	0
7	14	6	7	1	0	0	1	0	0	0
8	19	10	1	0	0	0	1	0	0	0
8	24	10	6	1	0	0	0	0	0	0
8	25	10	7	1	0	1	0	0	0	0
8	27	10	9	1	0	0	0	0	1	0
8	30	10	12	1	0	1	0	0	0	1
8	32	10	14	1	1	1	0	0	1	1
9	14	8	1	0	0	0	0	1	0	0
9	17	8	4	0	0	0	0	1	0	0
9	18	8	5	0	0	1	0	0	0	0
9	19	8	6	1	1	1	0	0	0	0
9	20	8	7	1	0	1	0	0	0	0
9	29	9	7	0	0	1	0	0	0	0
9	30	9	8	1	0	1	0	0	0	1

3.5 ECG Effects

- Some examples of ECG leads being removed (P2:A7, P8:A27)
- Accelerometer helps (P3:A6, P6:A7)
- Some of the 'subtle changes' observations may be associated with CVP pumped infusions.
- ECG noise in 9 plots
- ECG noise (sinusoidal) in 1 plot
- ECG "odd" in 2 plots

3.6 BP Effects

- (sinusoidal) on 1 plot

3.7 CVP Effects

- See P8:A25 for a very complex pattern. It would appear that the pump rate was slowed down mid annotation. Does this mean that they are doing a washing body procedure with a running IV medication procedure going on?

3.8 Accel Effects

- Often helps but phase shift issues. Could the phase shift be automatically corrected for particularly if there was some procedure when the accelerometer harness was fitted.
- Accelerometer scaling.
- 1 (chirp)
- 4 (non-chirp)

3.9 Most Common Combinations

- ECG and CVP noise combined

3.10 Non-Typical Presentations

- Sometimes you get something that just looks odd (P8:A32).
- No obvious artefact PID 9, ArtID 14 & ArtID 17

3.10.1 Misaligned Before Start

3.10.2 Misaligned After Start

3.11 Algorithm In Words

- Check for ECG leads disconnection.
- The presence of all three physiological *and* accelerometer signal disturbances could be used.
- step time < 1 minute (median/mean?), with "noise present on ECG and CVP signals or missing ECG even partially with noise on CVP signal

3.12 Clinical Questions

- How often is this type of procedure carried out when pumped IV infusions are ongoing?
- What is the most frequent part of body washed?
- Is the chest frequently washed and if so are the ECG leads typically disconnected?
- Does the body washing (chest, limbs, face?) usually occur in one go or are they staggered?

4 Blood Sample (N = 19, Number of Patients = 9)

Note the event list says we have 20. However due to the last artifact problem on patient 7 we only have 19.

4.1 Summary

Table 7: Artifact: Blood Sample Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Blood_sample	1	4	4	1	1 min 59.3 secs
	2	9	7	1	1 min 50 secs
	3	1	1	1	1 min 50 secs
	3	5	5	1	1 min 30 secs
	3	7	7	1	1 min 20 secs
	4	6	5	1	1 min 20 secs
	4	7	6	1	1 min 30 secs
	5	2	2	1	1 min 20 secs
	7	20	7	1	1 min 30 secs
	8	5	1	5	36.1 secs
	8	6	1	6	56.2 secs
	9	2	1	2	15.1 secs
	9	4	2	2	1 min 33.4 secs
	9	11	5	1	1 min 16.3 secs
	9	13	7	1	36.1 secs
	9	32	11	1	1 min 7.3 secs
	9	34	13	1	23.1 secs
	10	3	2	1	34.1 secs
	10	4	2	2	48.2 secs

4.2 Notes

Table 8: Artifact: Blood Sample Notes }

PID	ArtID	Task	Step	EPN	Note Text
1	4	4	1	-	1. This is a typical example of a blood sample. Note the large ”saturation” periods with Ramps and Zeroing fluctuations in the BP sample. 2. Saturation/Ramp/Zeroing components of the blood sample have been highlighted. 3. No changes of note in any other channel.

Table 8: *(Artifact: Blood Sample ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
1	4	4	1	1	1. This is a Zoomed in version of the previous blood sample (@fig-patient-1-Artifact-4) showing there are no obvious changes in the other channels. 2. Also demonstrates nicely the ectopics showing for patient 1 in the ECG trace.
2	9	7	1	-	1. Typical changes in BP (zero, ramp, saturation) during a blood sample
3	1	1	1	-	1. This is a typical blood sample with ramp, zero and saturation effects on the BP 2. Interestingly, later in the event, some noise on the BP channel and again later, some minor changes in accelerometer signals. Perhaps some manual handling of patient after the blood sample? 3. See the zoomed in plot below.
3	1	1	1	1	1. See the zoomed in plot, no effects on ECG but interesting noise pattern in BP, could this have been some sort of ”reverberation/resonance” after a line flush?
3	5	5	1	-	1. Again a typical example of a blood sample with ramp/zero/saturation elements in the BP signal.
3	7	7	1	-	1. A typical Blood sample event with ramp/zero/saturation features present in the BP signal.
4	6	5	1	-	1. Typical changes in BP signal (ramp, zero, saturation). 2. Also note there is some drift in the x,y planes of the accelerometer. Possibly not related to the blood sample, not seen this before. 3. Note the periodic pulses on CVP and other signals. Have seen this before with NG feeding... is it NG pump infusion artifact? After review it is most likely a pumped infusion pattern.
4	7	6	1	-	1. A text book blood sample event (ramp, zero, saturation) on BP signal. 2. No effects on other signals.
5	2	2	1	-	1. This is a typical Blood sample presentation (ramp, zero, saturation). 2. No effects on signals ECG or accelerometer. 3. Some artifact on CVP.
7	20	7	1	-	1. No effects seen. NOT a typical Blood sample event. Am thinking that either Kat missed it or it did not happen. Note the last event was also meant to be a blood sample, but it was off the end of the data-file. Somehow the last data file (patient7) was truncated short and I can’t see any reason why that might have been. So the last blood sample was lost.

Table 8: *(Artifact: Blood Sample ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
8	5	1	5	-	This plot shows the annotation period for this artifact does contain a ramp pattern toward the end. However looking at 1 minute of pre-annotation time suggests that the blood sample period may have started at the green cursor but can certainly be said to have started at the red cursor. The blue cursor is actually the start of the next artifact (6). This suggests that artifact 5 and 6 are part of *one* sequence of taking a blood sample. The CVP signal shows the pumped infusion pattern. There is observable PIIA on the ECG trace.
8	6	1	6	-	As described in the notes for artifact 5, this plot shows the characteristic zeroing and saturation sections of taking a blood sample. This artifact may be the end process of a normal blood sample procedure. The artifact seems to stop mid way through the annotation period at the green cursor. The CVP signal shows the pumped infusion pattern. There is observable PIIA on the ECG trace.
9	2	1	2	-	This is very odd in that there appears to be no artifact on any of the traces. I had a look at 1 and 2 minutes before and after and still no indication of a BP blood sample pattern.
9	4	2	2	-	Characteristic BP pattern. Substantial BP artifact just *after* the annotation end.
9	11	5	1	-	Flat line CVP at elevated level of 250 mmHg. No trace of the usual blood sample pattern for 20 minutes before or after the annotation time.
9	13	7	1	-	No obvious artifact. Where is the blood sample pattern? Checked pre and post for 10 mins.
9	32	11	1	-	BP signal shows characteristic blood sampling pattern. No obvious artifact on any of the other signals.
9	34	13	1	-	No obvious artifact on the ECG, BP or CVP traces. There is possibly some accelerometer movement across X, Y, Z and Sum signals during the annotation period and in the pre-annotation section. After a review this is unlikely to be associated with the blood sample procedure.

Table 8: *(Artifact: Blood Sample ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
10	3	2	1	-	This is the first step in a two step task. Both steps are blood samples. The annotation period shows the characteristic BP pattern however, it actually starts before the annotation period at the green cursor. The ECG signal shows some fluctuation but probably not artifactual. The CVP signal also shows some disturbance. There is a small amount of movement on the accelerometer Y signal. The red cursor actually shows the approximate start of the next artifact.
10	4	2	2	-	This is the end of the two step task. This annotation period includes the usual BP pattern for a blood sample however the procedure probably starts at the green cursor. The artifact period probably stops at the yellow cursor. Note the blood sample pattern before the green cursor is the previous artifact (ID 3). There is disturbance on the CVP signal. During the annotation period there is movement on the accelerometer X and Y signals.

4.3 Signal Summary

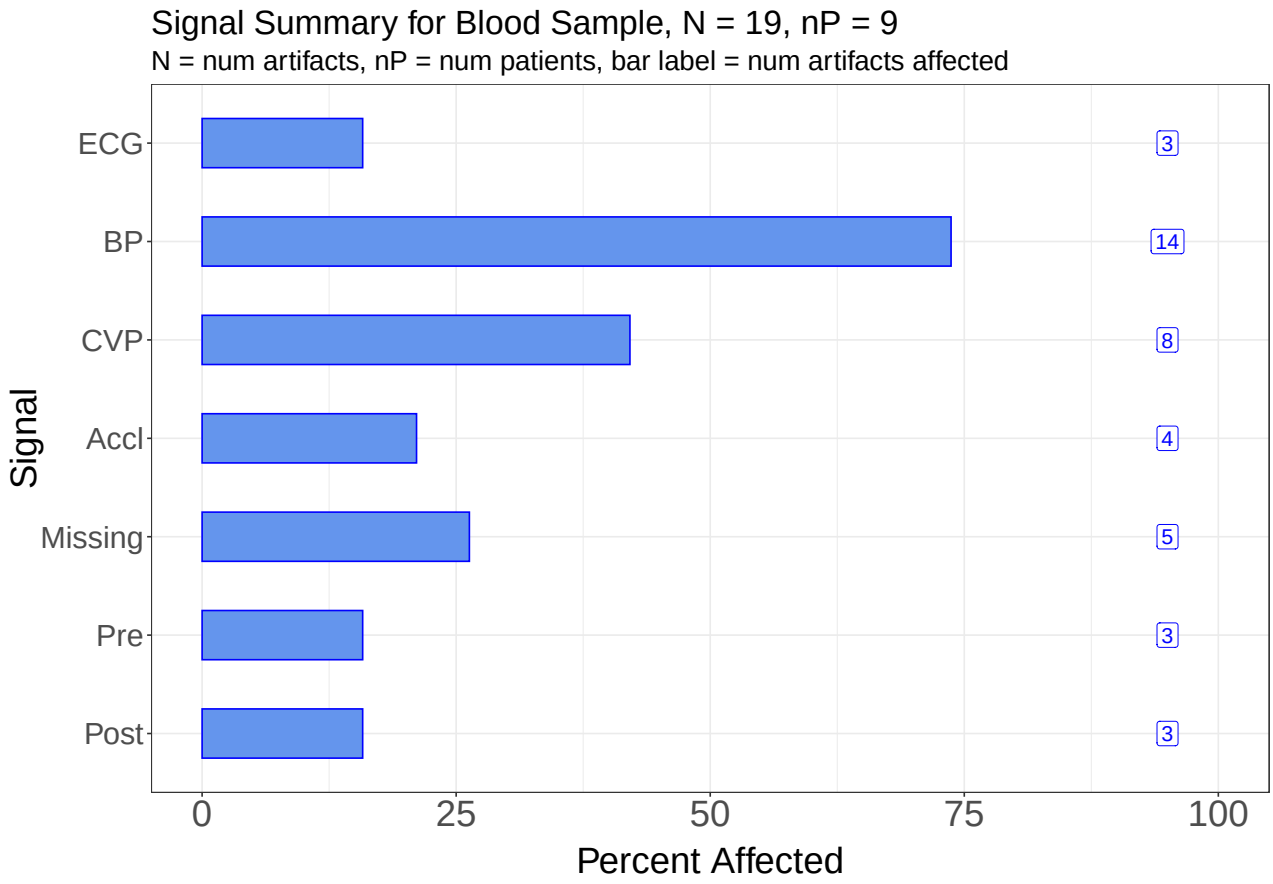

Figure 4: Artifact: Blood Sample Notes: Signal Summary

4.4 Notes Processing Summary

Table 9: Artifact: Blood Sample Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
1	4	4	1	0	1	0	0	0	0	0
2	9	7	1	0	1	1	0	0	0	0
3	1	1	1	1	1	1	1	0	0	0
3	5	5	1	0	1	0	0	0	0	0
3	7	7	1	1	1	1	0	0	0	0
4	6	5	1	0	1	1	0	0	0	0
4	7	6	1	0	1	0	0	0	0	0
5	2	2	1	0	1	1	0	0	0	0
7	20	7	1	0	0	0	0	1	0	0
8	5	1	5	0	1	0	0	0	1	0
8	6	1	6	0	1	0	0	0	0	0
9	2	1	2	0	0	0	0	1	0	0
9	4	2	2	1	1	1	1	0	0	1
9	11	5	1	0	0	0	0	1	0	0
9	13	7	1	0	0	0	0	1	0	0
9	32	11	1	0	1	0	0	0	0	0
9	34	13	1	0	0	0	0	1	0	0
10	3	2	1	0	1	1	1	0	1	1
10	4	2	2	0	1	1	1	0	1	1

4.5 ECG Effects

4.6 BP Effects

4.7 CVP Effects

- 1 flatline, probably not related

4.8 Accel Effects

- 1 occurred but during a non-typical event with no BP effects

4.9 Most Common Combinations

- BP changes typically with ramp, flush and saturation components with no other significant changes in the other signals.

4.10 Non-Typical Presentations

- P8:2, P9:13 and P7:20 no BP effects - did the actual event occur?

4.10.1 Misaligned Before Start

4.10.2 Misaligned After Start

4.11 Algorithm In Words

- step time < 2 minute (median/mean?)
- disruption in normal beat to beat of BP signal containing one or more ramp, zero and saturation features.

4.12 Clinical Questions

- Is there a SOP for drawing blood? or is there variation in approach? For example, is there always a flush of the line first before drawing blood back? Is there always expected to be a flush of the line after blood withdrawn?

5 Chest abdo limb exam (N = 14, Number of Patients = 7)

5.1 Summary

Table 10: Artifact: Chest abdo limb exam Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Chest_abdo_limb_exam	1	3	3	1	30 secs
	3	2	2	1	1 min 25 secs
	4	2	2	1	2 mins 10.5 secs
	6	1	1	1	45.2 secs
	6	14	4	1	1 min 0.2 secs
	7	5	3	1	10 secs
	7	6	4	1	10 secs
	8	1	1	1	1 min 30.4 secs
	8	2	1	2	39.2 secs
	8	4	1	4	6 secs
	8	13	7	1	45.2 secs
	8	14	7	2	8 secs
	8	15	7	3	18.1 secs
	9	5	3	1	58.2 secs

5.2 Notes

Table 11: Artifact: Chest_abdo_limb_exam Notes }

PID	ArtID	Task	Step	EPN	Note Text
1	3	3	1	-	1. This is an example of Chest-Abdo-Limb-Exam showing some change in the ECG baseline coupled with some change in the X-plane of the accelerometer.
3	2	2	1	-	1. Good example of patient handling during Chest-Abdo-Limb-Exam. Note the bursts of noise on the Accelerometer x,y,z planes. 2. Although the exact timing of the accelerometer noise do not match with the observed ECG/BP signal noise, they are likely causal. This demonstrates the existing phase lag between the time-clocks of the accelerometer and laptop real time clocks. 3. Also worth noting the different morphology of the x,y,z plane noise (more of a chirp shape) compared with a Turn event where there are sharp edges...

Table 11: *(Artifact: Chest Abdo Limb Exam ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
4	2	2	1	-	1. No obvious measurable effects on Physiology or Accelerometer. 2. CVP line is flat - suggest an infusion is in play.
6	1	1	1	-	1. Clear noise like changes in all channels that seem to sync temporarily. 2. CVP line sits at 250 mmHg suggesting an IV infusion going on.
6	14	4	1	-	1. Some clear effects of the limb exam on Accelerometer and frequent noise in the ECG. See the ep01 zoomed in plot below for detail.
6	14	4	1	1	1. See the noise on the ECG between green and red cursors.
7	5	3	1	-	1. No noticeable effects on any channels. Either Kat missed it, or the event effects were subtle.
7	6	4	1	-	No obvious artifact on ECG, CVP or Accelerometer X or Y. Possible artifact on the low pressure hump of the BP signal. Possible spikes on Accelerometer Z and Sum.
8	1	1	1	-	Possible artifact patterns on ECG, BP and CVP. The first (ECG only) at the green cursor, the second (ECG only) at the red cursor. The third area (ECG and CVP) is between the blue and yellow cursors. Interestingly there seems to be more artifact pre and post the annotation period which includes the BP signal as well as ECG and CVP. The accelerometer signals show little movement.
8	2	1	2	-	Considerable artifact on ECG, BP and CVP. The ECG trace shows some disturbance before the annotation start period at the green cursor. By the time at the red cursor (about half way through the annotation period) the artifact seems to have settled. The accelerometer signals show little movement apart from a cyclic pattern on the Y plane signal.
8	4	1	4	-	There does not seem to be any artifact within this annotation period. I checked +/- 0.5 mins and still no sign of any problems. The CVP signal seems 'choppy'. No movement on the accelerometer signals. A possible cyclic pattern on the Y plane signal.
8	13	7	1	-	The CVP signal is an example of a pumped infusion pattern.

Table 11: *(Artifact: Chest Abdo Limb Exam ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
8	14	7	2	-	There appears to be little artifact during the annotation period to the physiological signals however there is a disturbance on the accelerometer X signal. Adding +/- 0.25 mins to the plot shows artifact on the CVP signal and more movement on the accelerometer X signal. It looks like the procedure may well have occurred between the green and red cursors.
8	15	7	3	-	The CVP signal is an example of a pumped infusion pattern.
9	5	3	1	-	ECG artifact present.

5.3 Signal Summary

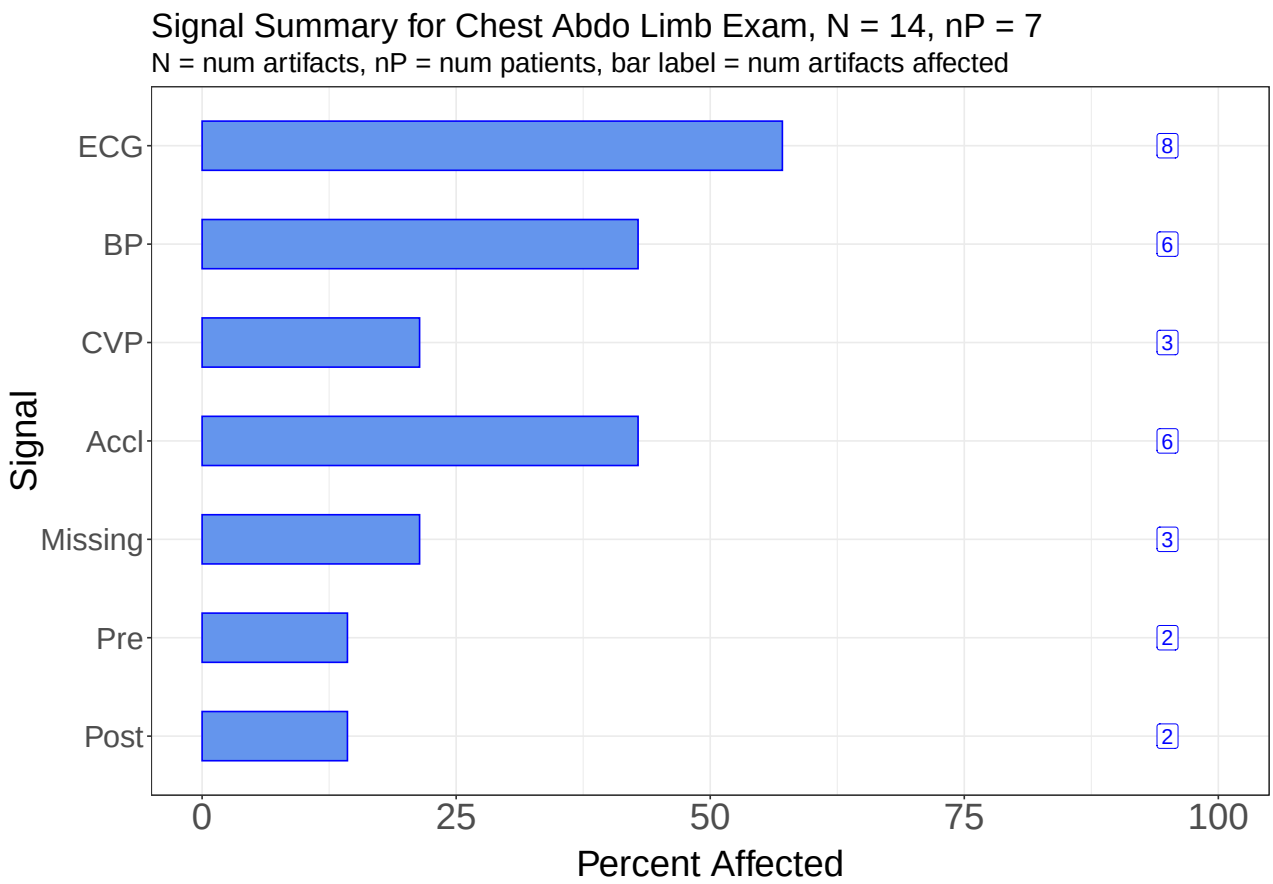

Figure 5: Artifact: Chest Abdo Limb Exam Notes: Signal Summary

5.4 Notes Processing Summary

Table 12: Artifact: Chest Abdo Limb Exam Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
1	3	3	1	1	1	0	1	0	0	0
3	2	2	1	1	1	1	1	0	0	0
4	2	2	1	0	0	0	0	1	0	0
6	1	1	1	1	1	0	1	0	0	0
6	14	4	1	1	0	0	1	0	0	0
7	5	3	1	0	0	0	0	1	0	0
7	6	4	1	0	1	0	1	0	0	0
8	1	1	1	1	1	1	0	0	1	1
8	2	1	2	1	1	1	0	0	1	0
8	4	1	4	0	0	0	0	1	0	0
8	13	7	1	1	0	0	0	0	0	0
8	14	7	2	0	0	0	0	0	0	1
8	15	7	3	0	0	0	1	0	0	0
9	5	3	1	1	0	0	0	0	0	0

5.5 ECG Effects

5.6 BP Effects

5.7 CVP Effects

- 2 (flat-line - infusion in play?)

5.8 Accel Effects

5.9 Most Common Combinations

- ECG then CVP, then Accel

5.10 Non-Typical Presentations

- PID4 ArtID 2 - no obvious Physiological or Accel noise
- PID7 ArtID 5 - no obvious Physiological or Accel noise
- PID8 ArtID 4 - no obvious Physiological or Accel noise

5.10.1 Misaligned Before Start

5.10.2 Misaligned After Start

5.11 Algorithm In Words

- Step time mostly less than 1 minute.
- Noise ECG with BP and frequently Accel signal too.

5.12 Clinical Questions

- Again - What is most frequent examination of the chest/abdomen/limb?
- Will they tend to all be done at once or staggered through the day?

6 Bed movement (N = 11, Number of Patients = 7)

6.1 Summary

Table 13: Artifact: Bed movement Notes Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Bed_movement	1	6	6	1	1 min 49.1 secs
	5	1	1	1	10 secs
	5	6	6	1	1 min 0 secs
	6	4	3	1	14.1 secs
	6	11	3	8	30 secs
	7	2	1	2	1 min 50 secs
	7	17	6	10	30 secs
	8	23	10	5	8 secs
	8	39	11	1	11 secs
	9	23	9	1	30.1 secs
	10	1	1	1	5 secs

6.2 Notes

Table 14: Artifact: Bed_movement Notes }

PID	ArtID	Task	Step	EPN	Note Text
1	6	6	1	-	1. NOT Typical Bed movement. No obvious changes in x,y,z planes, unless the bed movement was more of ”sheet adjustment” type as there was a lot of noise on the accelerometer signals. 2. Some change in CVP signal indicating that something is happening, but possibly not bed-movement?
5	1	1	1	-	1. Clear evidence of bed movement in the CVP and all three x,y,z planes of accelerometer. Also note this is one of the rare occasions where the Accel_Sum variable also shows a marked change. 2. The ”Chirp” shape of the noise profile suggests a burst or vibrational movement of the bed. This could happen when the head of the bed is raised or lowered as the motors/mechanics are not always ”smooth”.
5	6	6	1	-	1. Clear changes in all signals ECG, BP, CVP and accelerometer channels with bed movement.

Table 14: *(Artifact: Bed Movement ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
6	4	3	1	-	1. Fairly typical bed-movement event. Another example of a burst "Chirp" like signal on the CVP, ECG and BP signals whereas the Accelerometer changes shows more step like changes
6	4	3	1	1	1. Another bed-movement event but different character. More general noise throughout the bed-movement on all channels. Most change in y,z plane of Accelerometer.
6	11	3	8	-	1. No obvious effects of Bed-Movement. Had Kat missed this or perhaps it didn't happen?
7	2	1	2	-	1. Noise on ECG, CVP and BP signals (BP loses pulse in couple of occasions). 2. Usual step changes in x,y & z planes of accelerometer.
7	17	6	10	-	1. Bed movement detected by the x,y planes of the accelerometer. Only a very subtle effect on the ECG signal (5th ECG beat in from the left). No effect on BP signal.
8	23	10	5	-	This is an example of pumped infusion induced artifact. Two CVP spikes appear to be causing a disturbance on the ECG trace. There is no artifact on the BP signal.
8	39	11	1	-	No noticeable physiological signal artifacts. Possible accelerometer artifacts, mostly in the Z plane. Single CVP spike indicating that a pumped infusion is taking place.
9	23	9	1	-	This is the start of task 9, an 8 step procedure,ID 23 to 30. A possible CVP artifact between the red and green cursors (about 11:04). This is just before the annotation starts. Another possible artifact between the blue and yellow cursors. This is show in more detail in the next plot @fig-patient-9-Artifact-23-EP-01. The CVP does seem to shift level from about 10 mmHg to 15 to 20 mmHg over the course of the artifact annotation.
9	23	9	1	1	A possible artifact on a single blood pressure pulse.
10	1	1	1	-	This is the start of a two step task. Nothing obvious during the annotation period apart from the fact that the CVP signal seems to be missing and may well just be all artifact. Just after the annotation period there is some movement on the accelerometer signals. During the minute before the annotation period there is considerable disturbance on the ECG and BP signals. The CVP signal during this period seems to be all artifact.

6.3 Signal Summary

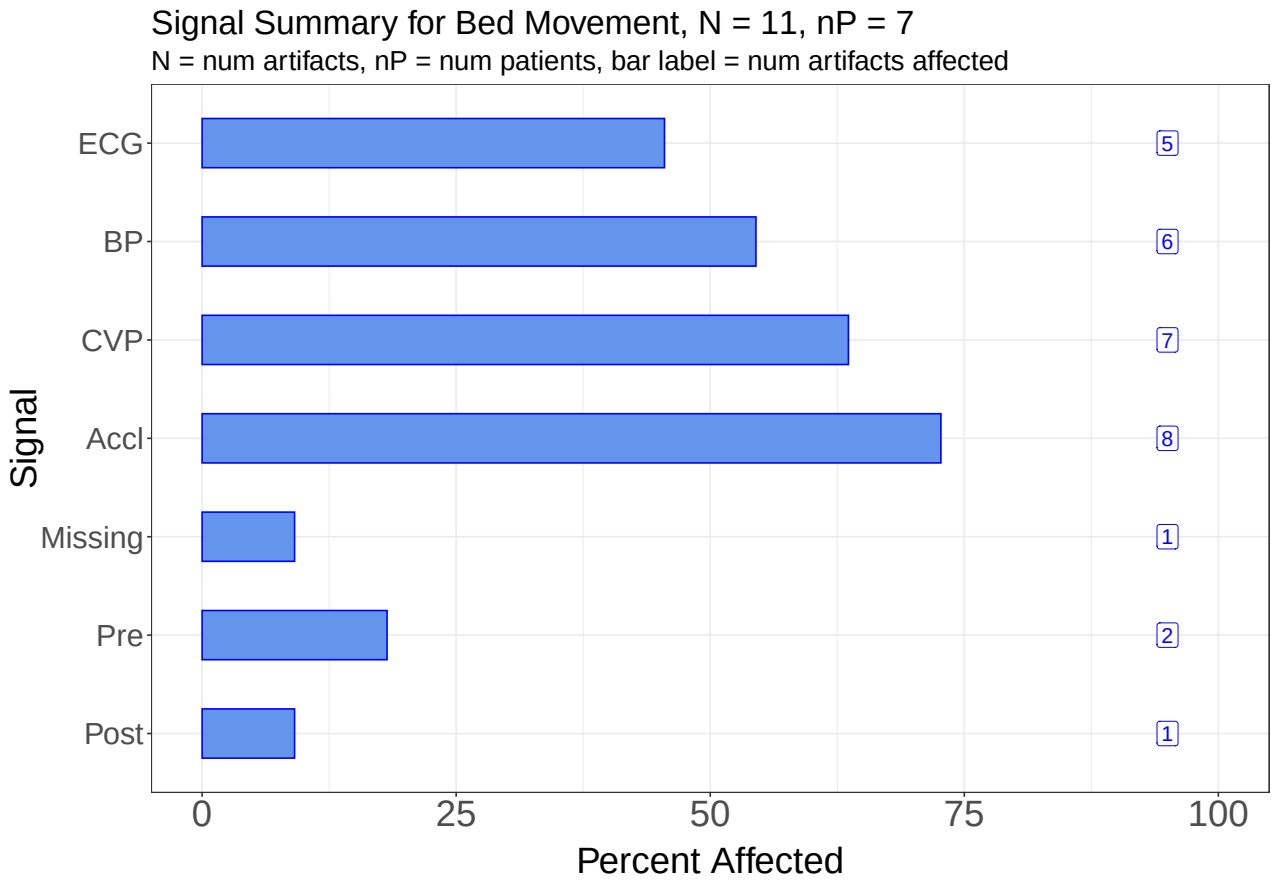

Figure 6: Artifact: Bed Movement Notes: Signal Summary

6.4 Notes Processing Summary

Table 15: Artifact: Bed movement Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
1	6	6	1	0	0	1	1	0	0	0
5	1	1	1	0	1	1	1	0	0	0
5	6	6	1	1	1	1	1	0	0	0
6	4	3	1	1	1	1	1	0	0	0
6	11	3	8	0	0	0	0	1	0	0
7	2	1	2	1	1	1	1	0	0	0
7	17	6	10	1	0	0	1	0	0	0
8	23	10	5	0	0	0	1	0	0	0
8	39	11	1	0	0	0	0	0	0	0
9	23	9	1	0	1	1	0	0	1	0
10	1	1	1	1	1	1	1	0	1	1

6.5 ECG Effects

- 1 (subtle)

6.6 BP Effects

6.7 CVP Effects

6.8 Accel Effects

- 3 (not typical expected noise - sheet movement?)
- 2 (chirp like signal…)
- 3 (most typical step like change)

6.9 Most Common Combinations

6.10 Non-Typical Presentations

- PID 6, ArtID 11 (no obvious changes - missed event?)

6.10.1 Misaligned Before Start

- considerable ECG and BP noise before annotation period (1 minute before)

6.10.2 Misaligned After Start

- CVP noise 2 seconds after annotation window
- accel noise just after annotation period

6.11 Algorithm In Words

6.12 Clinical Questions

7 Turn (N = 11, Number of Patients = 6)

7.1 Summary

Table 16: Artifact: Turn Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Turn	1	1	1	1	1 min 40 secs
	1	2	2	1	7 mins 10 secs
	1	5	5	1	1 min 55 secs
	2	8	6	1	6 mins 20.5 secs
	3	4	4	1	5 mins 40 secs
	4	5	4	1	9 mins 10 secs
	4	8	7	1	5 mins 40 secs
	6	13	3	10	15 secs
	7	1	1	1	19 mins 56.5 secs
	7	3	1	3	6 mins 50 secs
	7	18	6	11	6 mins 20 secs

7.2 Notes

Table 17: Artifact: Turn Notes }

PID	ArtID	Task	Step	EPN	Note Text
1	1	1	1	-	a) This Turn shows the typical changes in the X, Y & Z planes of the accelerometer signal indicating patient movement in all three planes during the turn. b) No obvious changes in ECG/BP associated with the turn, although CVP seems to have lost it's pulse which did not return even with extending the right hand window by nearly a minute. c) Note that there may seem to be some artifactual effects of turning on ECG and BP signals, although this patient has cardiovascular co-morbidites and is clearly having ectopic beats.
1	2	2	1	-	a) This turn event does not show a typical pattern. No obvious changes in ECG,BP or CVP signals associated with the turn. b) Also, no obvious changes in the accelerometer channels, which seem to be picking up some form of respiratory signal? c) This suggests that either the event did not happen or was missed by Kat?

Table 17: *(Artifact: Turn ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
1	5	5	1	-	1. This turn shows a typical example with clear changes in the x, y & z planes of the accelerometer. 2. There is also some noise appearing on the ECG and CVP channels which is not unexpected. 3. Note, due to some time lag between the accelerometer time-clock and the PC clock, movement induced changes may not be completely in sync but they should be within seconds of each other.
2	8	6	1	-	1. Clear typical changes in CVP and all accelerometer planes during a turn.
3	4	4	1	-	1. This is a typical example again of a turn event. Note the marked changes in x,y,z plane values. 2. There may be some influences of the turn on the ECG/BP signals, but they appear minor and non-specific thus underlining the effectiveness of the accelerometer signals for detecting turns.
4	5	4	1	-	1. Typical turn event with changes in x,y,z planes of the accelerometer. 2. Note some change in CVP signal too, not sure if this related to the turn. 3. Some noise on BP channel and note the temporary disruption in ECG electrode. Could be turn related.
4	8	7	1	-	1. NOT a normal turn profile. Some slow drift on y plane of accelerometer... could be consistent with a slow rolling of the patient onto side... but would expect at least z or x plane to change too. 2. No effects on ECG or BP.
6	13	3	10	-	1. Again, no signs of any effects on any channels... had Kat missed the turn or maybe it was postponed?
7	1	1	1	-	1. Significant noise and interruption of ECG and BP signals. CVP noise too with the usual step changes in Accelerometer x,y, z planes.
7	3	1	3	-	1. Marked step changes in x,y,z planes and also noise on CVP and ECG. 2. Note there is frequent ”Damping” of the BP trace. Probably not related to the Turn Event.
7	18	6	11	-	1. Clear effects of the turn (picked up by step changes in x,y planes of accelerometer). 2. Mostly BP signal affected with noise and ”damping” appearing... perhaps the turn has partially kinked the BP line?

7.3 Signal Summary

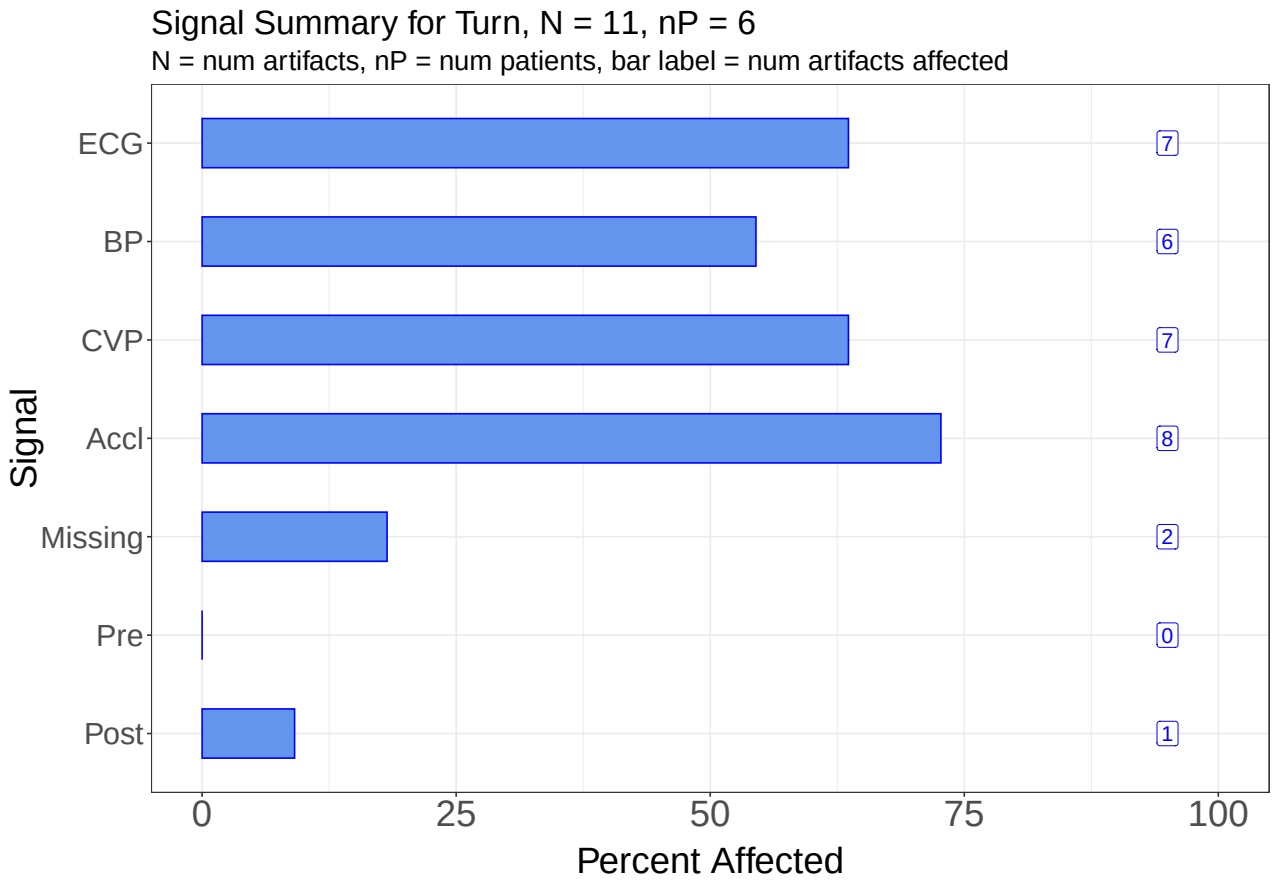

Figure 7: Artifact: Turn Notes: Signal Summary

7.4 Notes Processing Summary

Table 18: Artifact: Turn Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
1	1	1	1	0	0	1	1	0	0	1
1	2	2	1	0	0	0	0	1	0	0
1	5	5	1	1	0	1	1	0	0	0
2	8	6	1	1	1	1	1	0	0	0
3	4	4	1	1	1	1	1	0	0	0
4	5	4	1	1	1	1	1	0	0	0
4	8	7	1	0	0	0	0	1	0	0
6	13	3	10	1	0	0	0	0	0	0
7	1	1	1	1	1	1	1	0	0	0
7	3	1	3	1	1	1	1	0	0	0
7	18	6	11	0	1	0	1	0	0	0

7.5 ECG Effects

- 1 minor

7.6 BP Effects

- 2 minor

7.7 CVP Effects

7.8 Accel Effects

7.9 Most Common Combinations

- Accel changes dominate (step changes mostly), followed by CVP with a mixture of effects on ECG/BP

7.10 Non-Typical Presentations

- PID 1, ARTID 2, no obvious changes in any signal - missed?
- PID 4, ARTID 8, untypical slow drift changes - missed event?

7.10.1 Misaligned Before Start

7.10.2 Misaligned After Start

7.11 Algorithm In Words

- Typically "step changes" in one or more accelerometer x,y,z planes with more frequently effects on CVP and less frequently with ECG and BP.

7.12 Clinical Questions

- How often per nursing shift is the patient turned?
- Is it usually in one direction or is it variable?

8 Adjusting lines (N = 11, Number of Patients = 5)

8.1 Summary

Table 19: Artifact: Adjusting lines Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Adjusting_lines	4	3	3	1	40 secs
	6	3	2	1	25 secs
	6	15	5	1	2 mins 30 secs
	7	15	6	8	1 min 40 secs
	7	19	6	12	1 min 10 secs
	8	12	6	1	7 mins 5.6 secs
	9	9	3	5	11 secs
	9	10	4	1	8 secs
	9	16	8	3	2 secs
	9	22	8	9	14.1 secs
	9	28	9	6	18.1 secs

8.2 Notes

Table 20: Artifact: Adjusting_lines Notes }

PID	ArtID	Task	Step	EPN	Note Text
4	3	3	1	-	1. Possibly the CVP line was switched back onto the Xducer. Minor pulsation suggests this is the case rather than a Zeroing procedure. 2. No effects on Physiology or Accelerometer.
6	3	2	1	-	1. Not much to see... some variation in x,y plane of accelerometer and VERY subtle noise on ECG. See the extra EP01 plot for example.
6	3	2	1	1	1. In this zoomed in plot you can see the very subtle changes in the T-wave of the ECG and the BP pulse. This seems to be the only ”Noise like” that occurred. Doesn’t tie in though with the wave like pattern in the x,y planes of accelerometers suggesting that the Accel changes are probably unrelated to the line adjust event.
6	15	5	1	-	1. Considerable noise and signal disruption on all channels. Some accelerometer step change for the z plane too.
7	15	6	8	-	1. No obvious effects on any signals other than a slow upward drift in the x-plane of accelerometer, probably unrelated.

Table 20: *(Artifact: Adjusting Lines ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
7	19	6	12	-	1. Interestingly, clear signs of movement seen from x,y planes of accelerometer, but NO effects seen on ECG, BP or CVP signals.
8	12	6	1	-	CVP artifact at the start of annotation. Accelerometer movement at the end of the annotation.
9	9	3	5	-	Artifact present on CVP signal. Possible coincident artifact on CVP and BP signals about mid way through the annotation.
9	10	4	1	-	Straight line CVP signal at 250 mmHg (see the Y scale) compared to a usual value of about 15 mmHg. The CVP signal appears to have been lost about 10 minutes before the annotation start time. See the following plot
9	10	4	1	1	This CVP signal appears to have been lost about 10 minutes before the annotation start time at about 13:55. This is 9 minutes before the annotation starts but well after the end of the last artifact (Artifact 9, also an adjusting lines) at 13:35. You are going have to zoom in quite a bit to get clear view of the times.
9	10	4	1	2	This shows the times * [A] about 13:48 * [B] about 13:50 to 13:51 * [C] about 13:55, the 9 minute early start from extra plot 1, @fig-patient-9-Artifact-10-EP-01. Is this the adjusting lines procedure? This group is well after the last artifact @fig-patient-9-Artifact-9 which ends at 13:35.
9	16	8	3	-	No obvious artifact.
9	22	8	9	-	No obvious artifacts although there may be a hint of a CVP disturbance just after the green cursor. This is the end of Task 8 which started at Artifact 14.
9	28	9	6	-	Observable artifact at pre, during and post annotation period on BP and CVP. A possible movement on the Accelerometer X signal.

8.3 Signal Summary

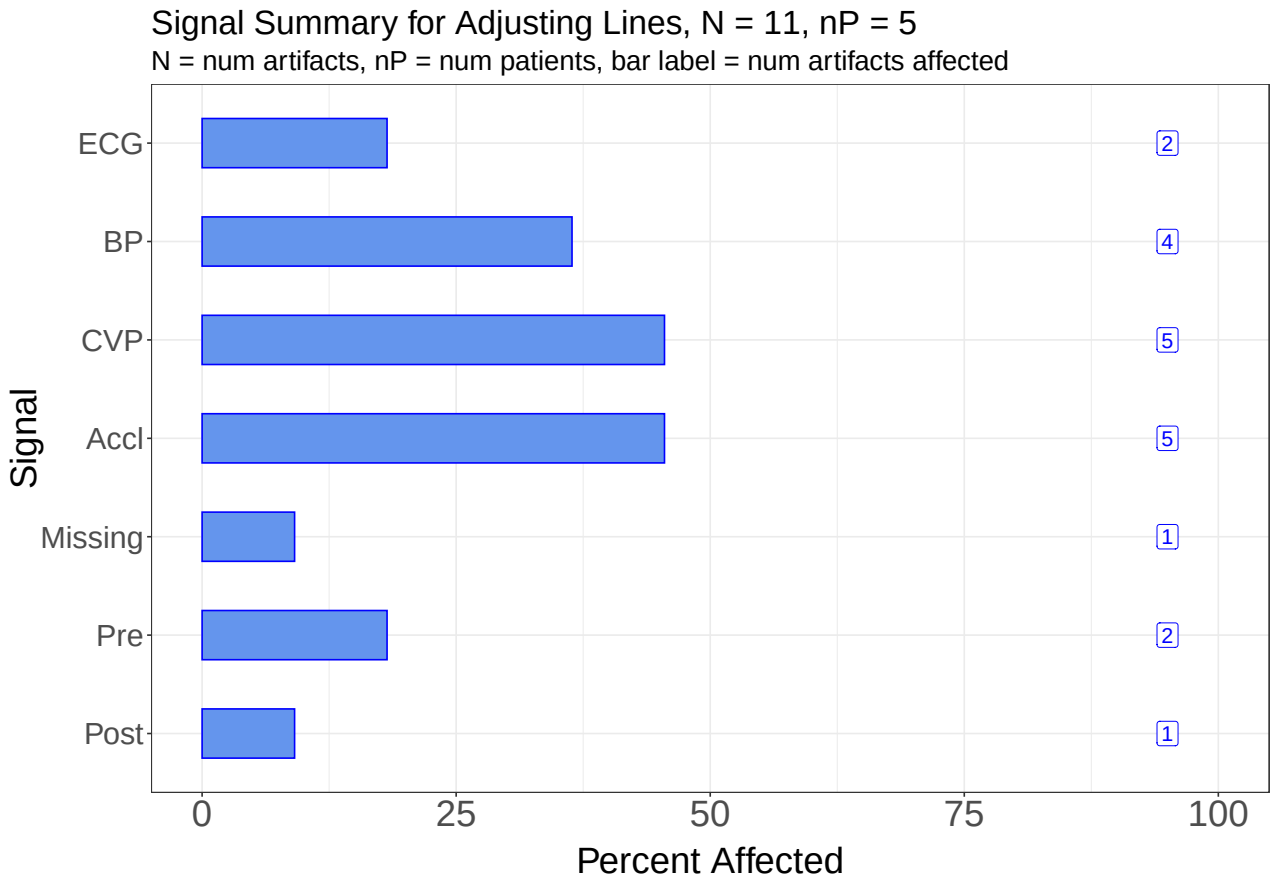

Figure 8: Artifact: Adjusting Lines Notes: Signal Summary

8.4 Notes Processing Summary

Table 21: Artifact: Adjusting Lines Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
4	3	3	1	0	0	0	0	0	0	0
6	3	2	1	1	1	0	1	0	0	0
6	15	5	1	1	1	1	1	0	0	0
7	15	6	8	0	0	0	0	0	0	0
7	19	6	12	0	0	0	1	0	0	0
8	12	6	1	0	0	1	1	0	0	0
9	9	3	5	0	1	1	0	0	0	0
9	10	4	1	0	0	0	0	0	1	0
9	16	8	3	0	0	0	0	1	0	0
9	22	8	9	0	0	1	0	0	0	0
9	28	9	6	0	1	1	1	0	1	1

8.5 ECG Effects

- 2 subtle noise on ECG

8.6 BP Effects

- 4 subtle noise on BP

8.7 CVP Effects

- 1 CVP line switched back on no effects on other channels.

8.8 Accel Effects

8.9 Most Common Combinations

- BP and CVP effects most, occasional accel effects. Often subtle. No consistent features jumping out.

8.10 Non-Typical Presentations

8.10.1 Misaligned Before Start

- 1 CVP line change before start.

8.10.2 Misaligned After Start

8.11 Algorithm In Words

- Step time less than 2 minutes, subtle changes in some signals without any consistent "linked Pattern" between ECG, BP and CVP (which would suggest a physiological response). Most likely feature to trigger on would be CVP line pulse appearing/disappearing compared to a flat line (near zero) or a saturated line (about 250 mmHg) suggestive of CVP line infusion.

8.12 Clinical Questions

- Annotation approach for this type of artifact is probably not good enough to separate between line adjustments. Be good to chat with Kat to discuss what types of events were coded as line-adjustments.

9 Gown change (N = 11, Number of Patients = 4)

9.1 Summary

Table 22: Artifact: Gown change Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Gown_change	6	5	3	2	1 min 8.3 secs
	6	10	3	7	2 mins 5.3 secs
	7	8	6	1	2 mins 30 secs
	7	9	6	2	55 secs
	7	16	6	9	2 mins 15 secs
	8	21	10	3	40.2 secs
	8	34	10	16	1 min 22.3 secs
	9	15	8	2	1 min 15.3 secs
	9	21	8	8	1 min 56.4 secs
	9	25	9	3	17.1 secs
	9	27	9	5	29.1 secs

9.2 Notes

Table 23: Artifact: Gown_change Notes }

PID	ArtID	Task	Step	EPN	Note Text
6	5	3	2	-	1. Clear noise in the ECG, BP, CVP signals with gown change. 2. Note the accelerometer signal changes are more variable probably as a gown change event moves the patient limbs/body in a more variable way.
6	10	3	7	-	1. No obvious effects on any channels - has Kat missed the gown change?
7	8	6	1	-	1. No clear changes in any signals to suggest there was manipulation during a gown change. 2. Very brief change in baseline of ECG but no noticeable x,y,z changes which you would expect during moving patients limbs during a gown change.
7	9	6	2	-	1. Similarly, no noticeable effects on any channels. 2. Is Kat missing these events or timing is off or they were cancelled?
7	16	6	9	-	1. This looks like limb/chest movement being picked up and ECG and BP noise related to noise in the x,y,z planes of the accelerometer.

Table 23: *(Artifact: Gown Change ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
8	21	10	3	-	An odd pulsing pattern on the CVP signal that starts well before the annotation period and continues after it. The annotation period shows artifact patterns on the ECG and BP signals. In the 0.5 mins before the annotation start you can see at the green cursor when the procedure may be starting. The procedure may be ending at the blue cursor, which is past the annotation end. There is quite a stable period between the red (mid annotation period) and blue cursors. The X plane of the accelerometer shows some disturbance pre and post annotation period. The other accelerometer signals do not show any obvious artifact pattern.
8	34	10	16	-	The CVP signal is showing the pumped infusion pattern. Some possible ECG artifact. Possible BP artifact lagging ECG artifact.
9	15	8	2	-	Artifact on BP signal.
9	21	8	8	-	Multiple artifacts across the physiological signals. Nothing obvious on the accelerometer signals.
9	25	9	3	-	Possible small artifact on CVP (red cursor), BP (blue cursor) and CVP (yellow cursor). However there is considerable artifact in the pre-annotation period on both CVP and BP signals. The green cursor is positioned at the end of the last artifact annotation time. There is no noticeable change in the accelerometers signals which may suggest that the accelerometer device was removed for the gown change procedure.
9	27	9	5	-	Obvious artifact patterns on CVP and BP signals. The artifact on the BP signal extends beyond the annotation end. The ECG signal shows some disturbance, whether this is an actual physiologic response or artifact would need further investigation. No artifact pattern on the accelerometer signals. Again I would have thought the accelerometer needs to be removed for the gown change.

9.3 Signal Summary

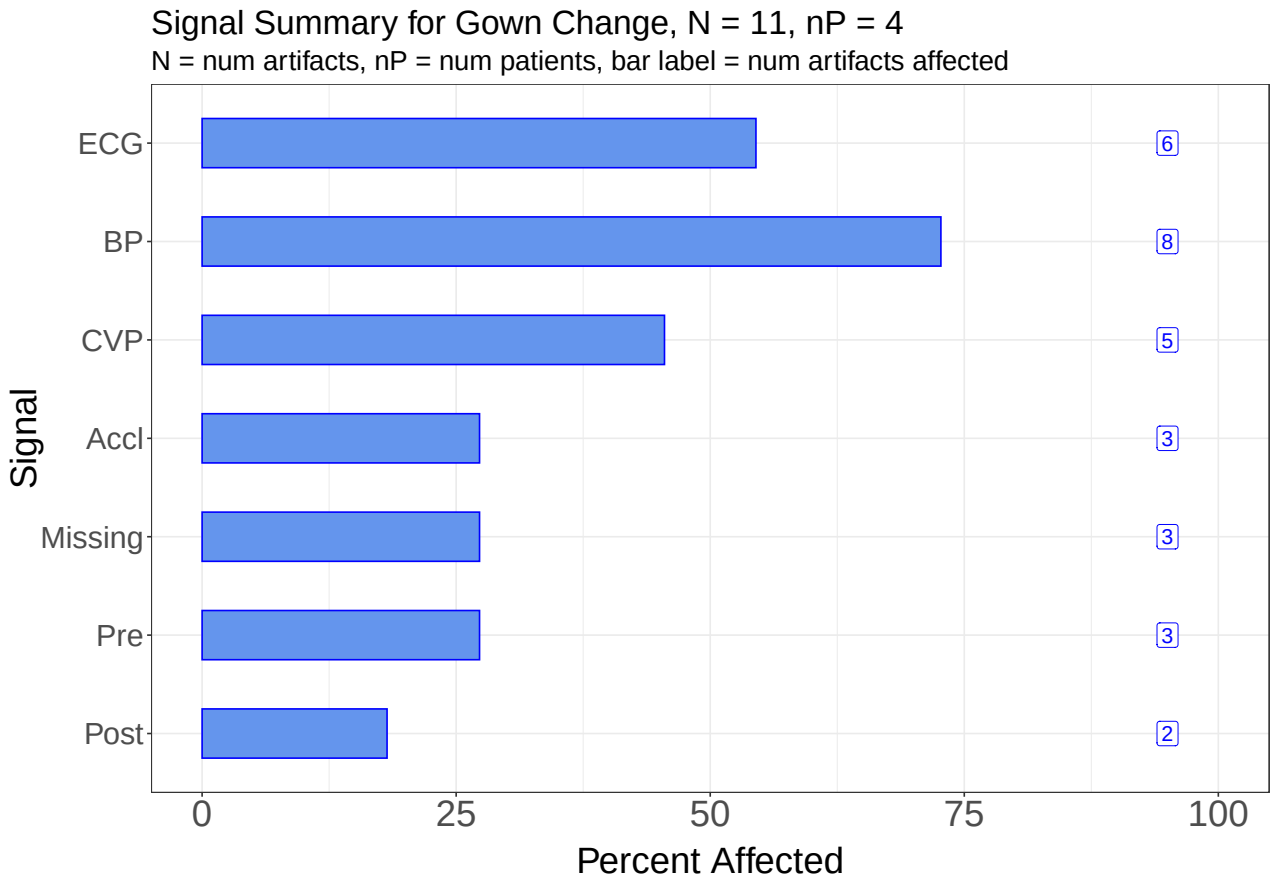

Figure 9: Artifact: Gown Change Notes: Signal Summary

9.4 Notes Processing Summary

Table 24: Artifact: Gown Change Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
6	5	3	2	1	1	1	1	0	0	0
6	10	3	7	0	0	0	0	1	0	0
7	8	6	1	0	0	0	0	1	0	0
7	9	6	2	0	0	0	0	1	0	0
7	16	6	9	1	1	0	1	0	0	0
8	21	10	3	1	1	1	1	0	1	1
8	34	10	16	1	1	0	0	0	1	0
9	15	8	2	0	1	0	0	0	0	0
9	21	8	8	1	1	1	0	0	0	0
9	25	9	3	0	1	1	0	0	1	0
9	27	9	5	1	1	1	0	0	0	1

9.5 ECG Effects

9.6 BP Effects

9.7 CVP Effects

9.8 Accel Effects

- Accelerometer effects low at about 25%. This makes sense to me in that doesn't the accelerometer harness need to be taken off for a gown change? Given that, why is there even 25%?

9.9 Most Common Combinations

- Most common, ECG, BP and CVP. Accel changes less frequent.

9.10 Non-Typical Presentations

- PID 6, ArtID 10 - missed event?
- PID 7, ArtID 8 - missed event?
- PID 7, ArtID 9 - missed event?

9.10.1 Misaligned Before Start

- 1 (PID 8, ArtID 21)
- 1 (PID 9, ArtID 25)

9.10.2 Misaligned After Start

9.11 Algorithm In Words

- Noise in physiological signals which less likely to be mirrored in accelerometer signals.

9.12 Clinical Questions

- See 'Accel Effects', procedure for gown change?
- Gown change, how often? Is there a technique? ECG electrodes disconnected?

10 IV_medication (N = 11, Number of Patients = 4)

10.1 Summary

Table 25: Artifact: IV medication Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
IV_medication	2	1	1	1	5 mins 40 secs
	2	2	2	1	42.2 secs
	2	3	3	1	5 mins 50 secs
	2	4	3	2	6 mins 40 secs
	2	5	3	3	1 min 10 secs
	2	6	4	1	2 mins 59.7 secs
	4	1	1	1	1 min 30 secs
	4	4	3	2	50 secs
	5	4	4	1	6 mins 27.5 secs
	7	4	2	1	3 mins 53.8 secs
	7	7	5	1	2 mins 7.1 secs

10.2 Notes

Table 26: Artifact: IV_medication Notes }

PID	ArtID	Task	Step	EPN	Note Text
2	1	1	1	-	1. Note the saturation of the CVP line. As expected with medication bolus or infusion. 2. Note the minor changes in x,y,z accelerometer planes. Don't know what this means. 3. Note also some effects on BP and ECG. Small increase in BP might just be fluid loading through central line. Minor changes in ECG might be a similar effect. See the detailed plot below.
2	1	1	1	1	1. At the green-cursor is the start of the effects on BP and ECG. Most marked changes in BP at the red cursor. 2. No obvious change in heart-rate during the infusion, but the ECG baseline did change.
2	2	2	1	-	1. No clear changes seen here with the "medication" 2. Hint that CVP pattern changed but it is periodic anyway so unlikely due to the medication.
2	3	3	1	-	1. Again, no obvious changes in ECG, BP or Accelerometers to do with Medication.

Table 26: *(Artifact: IV Medication ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
2	4	3	2	-	1. CVP changes consistent with infusion/bolus of medication through the central line. 2. No obvious changes in ECG, BP signals 3. Possibly some very subtle changes in the accelerometer signals though. Need to ask Annemarie about this but it may be the central line 3-way tap they use for bolus injection or infusion may sit on the chest or neck and possibly explain some subtle chest movement when injecting...
2	5	3	3	-	1. Another medication. This time there is a clear noise element on the CVP signal which projects to the BP signal but not the ECG channel. 2. Some noise episodes in the Accelerometer signal too possibly explained by the potential movement of 3-way tap on chest?
2	6	4	1	-	1. Medication effects on CVP, BP and X plane of accelerometer. See zoomed in plots below.
2	6	4	1	1	1. See changes in CVP and some BP effects between Green and Red cursors
2	6	4	1	2	1. This is a Zoomed in section showing the medication which caused noise on the CVP and BP channels had no effect on ECG.
4	1	1	1	-	1. Clear noise on the CVP channel, no effects on ECG/BP/Accelerometer 2. Is this a medication or some sort of handling of the CVP line (or both?)
4	4	3	2	-	1. Clear noise/changes in CVP signal. I suggest some manipulation of the CVP line 3-way tap followed by a couple of bolus's. 2. No Effects on BP/ECG or Accelerometer.
5	4	4	1	-	1. No clear effects of medication. A couple of abnormalities in ECG and one which mapped to BP as well. See the two following extra plots for details. 2. A minor fluctuation in the y-plane of the accelerometer, probably not related.
5	4	4	1	1	1. Note between the Green and Red cursors. There was a change in ECG baseline, but no clear ECG abnormality other than the shift in baseline. No clear evidence this is related to the medication event.
5	4	4	1	2	1. On Zooming in, although there seemed to be some sort of change in ECG/BP "density" but this didn't translate into any features of the signals suggesting abnormality.

Table 26: *(Artifact: IV Medication ... continued)*

PID	ArtID	Task	Step	EPN	Note Text
7	4	2	1	-	1. Some noise on CVP. Some small step change in x,y planes of accelerometer 2. Note there is frequent "Damping" of the BP trace. Probably not related to the IV Medication.
7	7	5	1	-	1. Clear noise event in CVP suggesting a bolus? Which was followed by noise on ECG and BP at two different times over the event. 2. x,y changes in accelerometer... suggesting movement of the chest during bolus injection on the CVP line which is probably on or near the chest?

10.3 Signal Summary

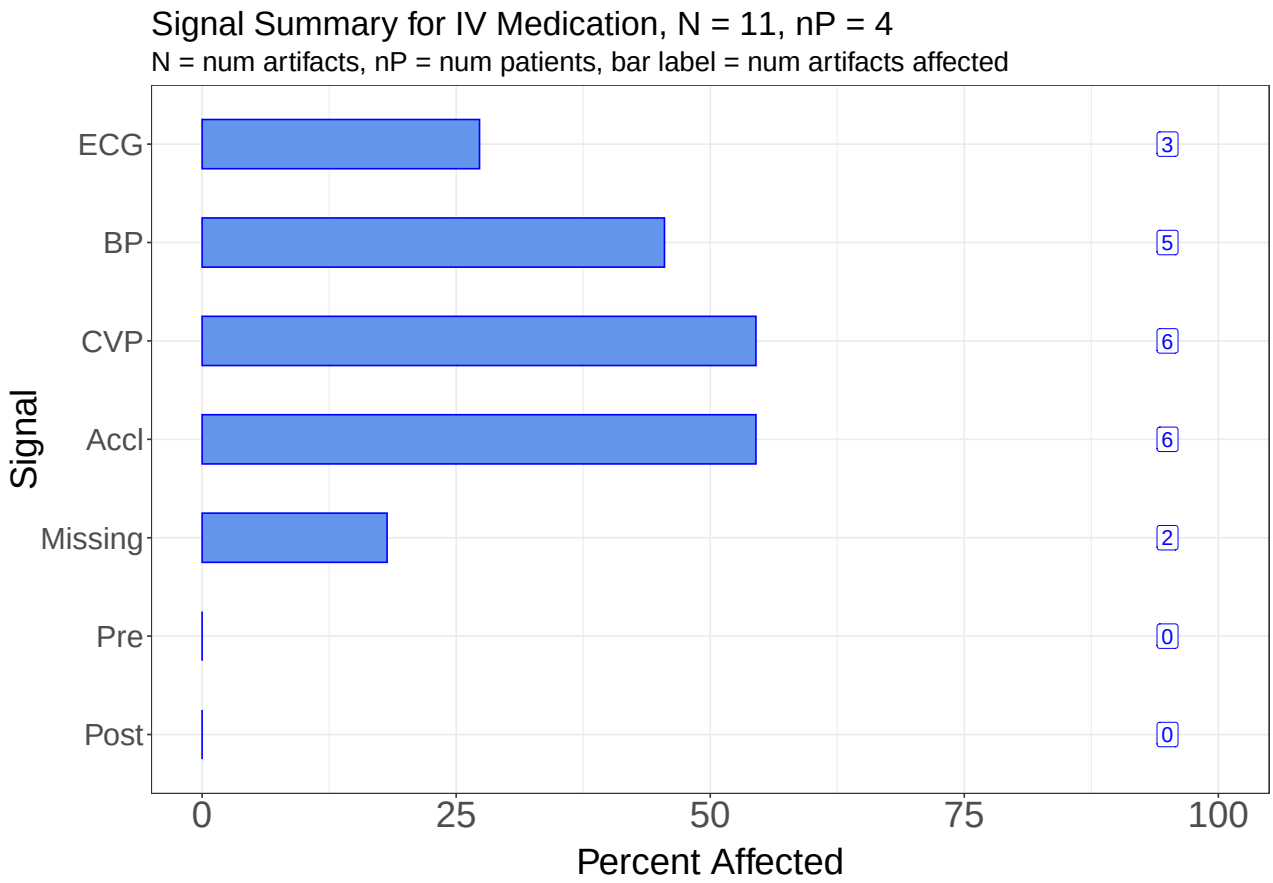

Figure 10: Artifact: IV Medication Notes: Signal Summary

10.4 Notes Processing Summary

Table 27: Artifact: IV Medication Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
2	1	1	1	1	1	0	1	0	0	0
2	2	2	1	0	0	0	0	1	0	0
2	3	3	1	0	0	0	0	1	0	0
2	4	3	2	0	0	0	1	0	0	0
2	5	3	3	0	1	1	1	0	0	0
2	6	4	1	0	1	1	1	0	0	0
4	1	1	1	0	0	1	0	0	0	0
4	4	3	2	0	0	1	0	0	0	0
5	4	4	1	1	0	0	0	0	0	0
7	4	2	1	0	1	1	1	0	0	0
7	7	5	1	1	1	1	1	0	0	0

10.5 ECG Effects

- 2 (minor)

10.6 BP Effects

- 1 (minor)

10.7 CVP Effects

10.8 Accel Effects

- 1 subtle

10.9 Most Common Combinations

- CVP and BP with Accel

10.10 Non-Typical Presentations

- PID 2, ArtID 2 - no obvious changes
- PID 2, ArtID 3 - no obvious changes

10.10.1 Misaligned Before Start

10.10.2 Misaligned After Start

10.11 Algorithm In Words

- Noise or "flattened/saturated" CVP (most common) with noise also on BP and sometimes accelerometer.

10.12 Clinical Questions

- IV bolus vs Infusion, assume losing the CVP signal and it being saturated about 250 mmHg is indicating a pressure infusion through the CVP line. Is the CVP transducer switched off to the infusion sometimes?

11 ECG lead (N = 9, Number of Patients = 3)

11.1 Summary

Table 28: Artifact: ECG lead Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
ECG_lead	6	9	3	6	2 mins 20.3 secs
	6	12	3	9	50 secs
	7	12	6	5	52.2 secs
	7	13	6	6	1 min 15 secs
	8	26	10	8	30.1 secs
	8	28	10	10	27.1 secs
	8	29	10	11	12.1 secs
	8	31	10	13	1 min 7.3 secs
	8	33	10	15	38.2 secs

11.2 Notes

Table 29: Artifact: ECG Lead Notes }

PID	ArtID	Task	Step	EPN	Note Text
6	9	3	6	-	1. Not much noise apart from a minor amount in the ECG trace. See the Zoomed in trace below
6	9	3	6	1	1. Between the green and red cursors one can see some ECG noise. Possibly adjusting the ECG electrodes. Note no effects on BP. CVP has a lot of spike like noise on it but no indication it is anything to do with the ecg_lead event.
6	9	3	6	2	1. Clean example of only ECG and CVP channels effected during lead event.
6	12	3	9	-	1. No obvious ECG Lead event, unless it was the short swell in baseline at the beginning of the event?
7	12	6	5	-	1. Some minor noise in the ECG signal that ties in with a delayed change in x,y accelerometer signal.
7	12	6	5	1	1. Unclear changes on this ECG lead event. Two extra beats in ECG... is that the effects of jostling the lead or is it physiological?
7	13	6	6	-	1. More marked change on BP signal rather than ECG signal. Some movement on x,y plane suggests this involves pressure/movement of the chest?

Table 29: *(Artifact: ECG Lead … continued)*

PID	ArtID	Task	Step	EPN	Note Text
8	26	10	8	-	Annotation timing seems off. Added 30 secs before and 60 secs after to pick up the obvious start and end pattern. On reflection this shows the difficultly of manual timing as the whole procedure looks like it took 1 min 45 secs (3322321 - 3301322) * 5 ms.
8	28	10	10	-	This is a bit odd. There is a definite ECG artifact at the start of the annotation but it seems to be at the end of an ECG break. Added 1.8 mins to the start to spot the start of the ECG break. I have added two ECG plots as extra information. The first ECG plot starts at 3321500 which is 1410 ms *before* the annotation start time [(3321782 - 3321500) * 5 ms]. You can see what I imagine is the reconnection of the ECG leads. This is followed at least two odd ECG patterns as the system settles. The second ECG plot starts 8 seconds later at 3323100 [(3323100 - 3321500) * 5 ms] to give a 2 second overlap with the first ECG plot. You can see by 3 seconds the ECG pattern seems to have settled.
8	28	10	10	1	This ECG plot starts at 3321500 which is 1410 ms *before* the annotation start time [(3321782 - 3321500) * 5 ms]. You can see watch I imagine is the reconnection of the ECG leads. This is followed at least two odd ECG patterns as the system settles.
8	28	10	10	2	This ECG plot starts 8 seconds after the previous ECG plot at 3323100 [(3323100 - 3321500) * 5 ms] to give a 2 second overlap with the first ECG plot. You can see by 3 seconds the ECG pattern seems to have settled.
8	29	10	11	-	Single ECG artifact starting about 3329452. A delayed CVP artifact starting at 3329804.
8	31	10	13	-	Artifact annotation appears to start too early and finish too soon. The CVP signal is showing the pumped infusion pattern.
8	33	10	15	-	Artifact annotation seems mistimed. The start might be OK but the end time seems to be well after things have settled albeit the *well after* is only 28.4 secs [(3370972 - 3365292) * 5 ms] The CVP signal is showing the pumped infusion pattern.

11.3 Signal Summary

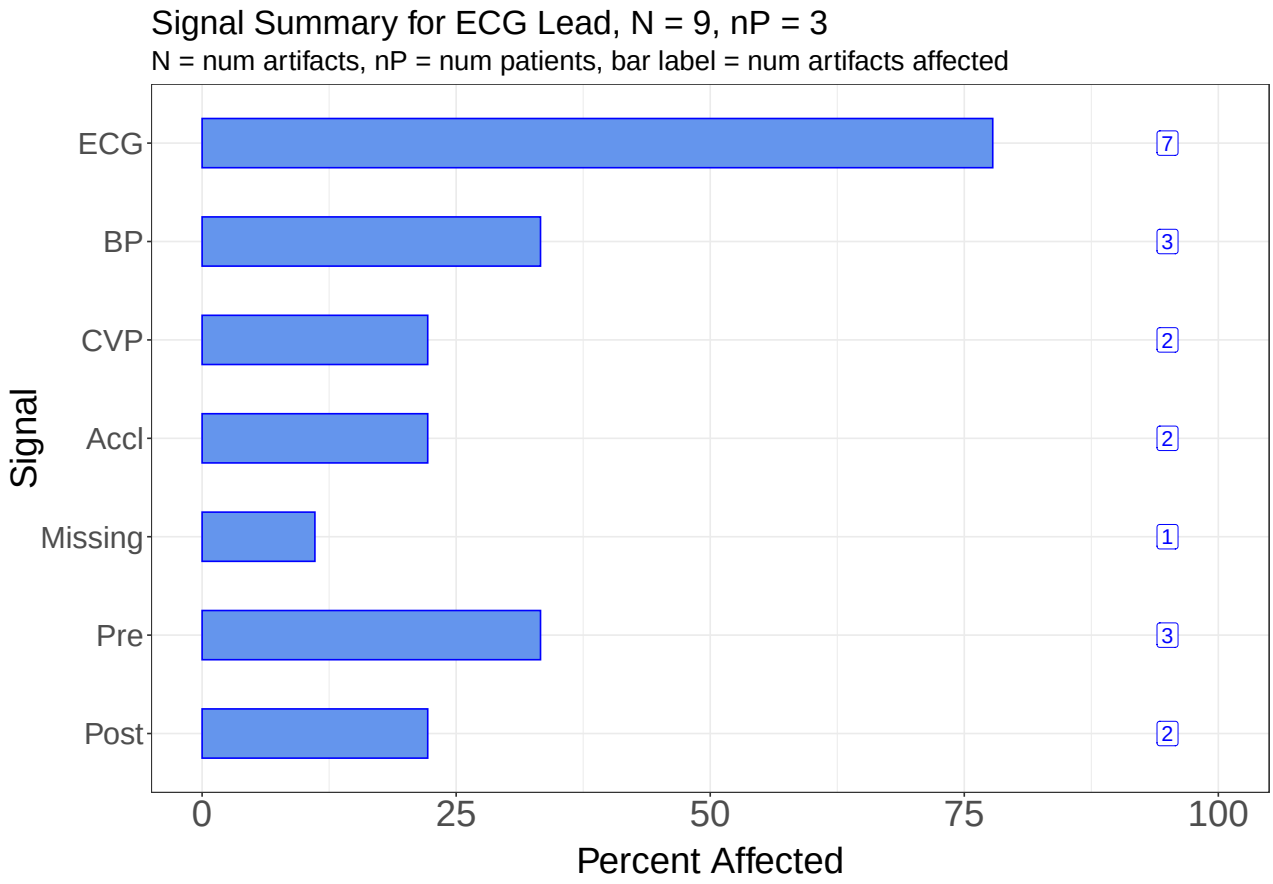

Figure 11: Artifact: ECG Lead Notes: Signal Summary

11.4 Notes Processing Summary

Table 30: Artifact: ECG Lead Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
6	9	3	6	1	0	1	0	0	0	0
6	12	3	9	0	0	0	0	1	0	0
7	12	6	5	1	0	0	1	0	0	0
7	13	6	6	0	1	0	1	0	0	0
8	26	10	8	1	0	0	0	0	1	1
8	28	10	10	1	0	0	0	0	1	0
8	29	10	11	1	0	1	0	0	0	0
8	31	10	13	1	1	0	0	0	0	0
8	33	10	15	1	1	0	0	0	1	1

11.5 ECG Effects

- 1 minor (altered beats? Physiological?)

11.6 BP Effects

11.7 CVP Effects

11.8 Accel Effects

11.9 Most Common Combinations

- ECG effects mostly with occasional CVP and Accel effects. Often the window is not accurate, starting and ending beyond the window.

11.10 Non-Typical Presentations

- PID 6, ArtID 12 - Missed?

11.10.1 Misaligned Before Start

- PID 8, ArtID 26 and ArtID 28 - ECG effect before and after window

11.10.2 Misaligned After Start

- PID 8, ArtID 26 - ECG effect before and after window
- PID 8, ArtID 33 - Delayed ECG effect after

11.11 Algorithm In Words

- look for ECG signal disruption/noise with CVP or accel accompanying to be confident it is ECG lead adjustment. May need an expanded window.

11.12 Clinical Questions

- again, how often are the ECG lead adjusted? Is it usually associated with turning? PAC?

12 Pupils (N = 7, Number of Patients = 2)

12.1 Summary

Table 31: Artifact: Pupils Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Pupils	8	10	5	1	32.1 secs
	8	11	5	2	11 secs
	8	17	9	1	11 secs
	8	18	9	2	9 secs
	9	3	2	1	24.1 secs
	9	12	6	1	18.1 secs
	9	33	12	1	19.1 secs

12.2 Notes

Table 32: Artifact: Pupils Notes }

PID	ArtID	Task	Step	EPN	Note Text
8	10	5	1	-	The annotation period appears to contain no artifact. However adding 0.5 mins pre-annotation time shows two artifacts on the BP trace (red and green cursors), see the next plot @fig-patient-8-Artifact-10-EP-01. There is a pulsing pattern on the CVP signal which indicates there is a pumped infusion taking place.
8	10	5	1	1	This plot zooms in on the 0.5 mins pre-annotation time and shows two artifacts on the BP trace (red and green cursors). The pulsing pattern on the CVP signal is more distinct.
8	11	5	2	-	Marked pattern on CVP which extends just beyond end annotation time. This is an example of a pumped infusion pattern.
8	17	9	1	-	The CVP signal shows a pumped infusion pattern.
8	18	9	2	-	Pumped infusion pattern in CVP showing typical spikes. These spikes also seem to coincide with small spikes on the ECG (red ovals). After discussion with clinical colleagues it is agreed that this is an example of pumped infusion induced artifact (PIIA) most likely caused by the setup of the infusion tubing/system.

Table 32: *(Artifact: Pupils … continued)*

PID	ArtID	Task	Step	EPN	Note Text
8	18	9	2	1	This is the left hand spike from @fig-patient-8-Artifact-18 discussed above. Notice that there is no artifact on the BP signal. This fact would help the ICU team in deciding whether or not this was an artifact.
8	18	9	2	2	This is the right hand spike from @fig-patient-8-Artifact-18 discussed above. Again notice that there is no artifact on the BP signal.
8	18	9	2	3	The ECG 10 second rhythm strip covering the two CVP spikes from the pumped infusion induced artifact (PIIA). From @fig-patient-8-Artifact-18, the first CVP spike was at sample number 3065317, the second at sample number 3066623. This is 1306 * 5 ms = 6530 ms. This looks like it matches the yellow ovals on the above ECG trace.
9	3	2	1	-	No noticeable artifact patterns on any signal.
9	12	6	1	-	No obvious artifact.
9	33	12	1	-	There is one notable section of artifact on the three physiologic signals within the annotation period. However there are several disturbance patterns across ECG, BP and CVP in the approximately two minutes before the annotation period starts (from the green cursor). I wonder if this is artifact or a physical response to the pupils check procedure. Nothing apparent on the accelerometer signals.

12.3 Signal Summary

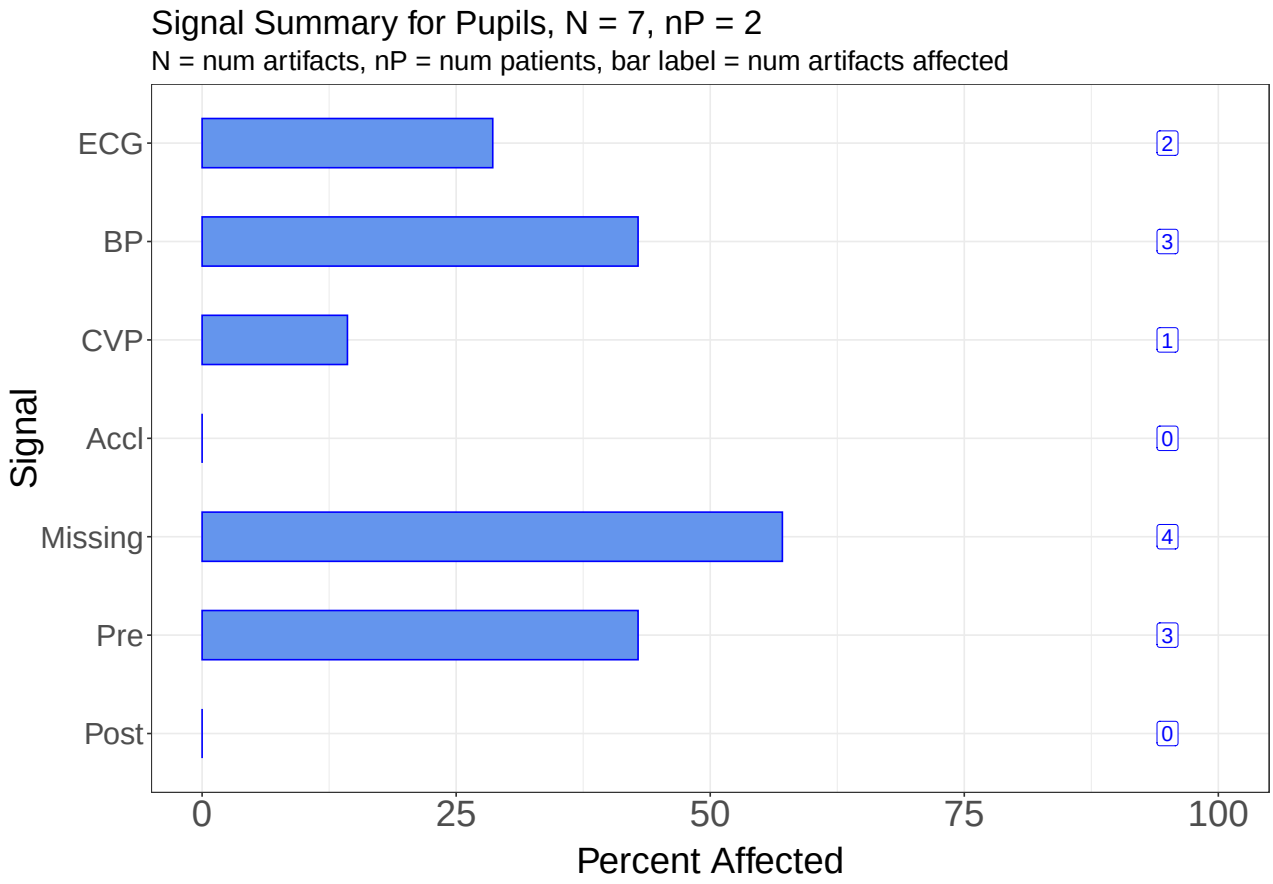

Figure 12: Artifact: Pupils Notes: Signal Summary

12.4 Notes Processing Summary

Table 33: Artifact: Pupils Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
8	10	5	1	0	1	0	0	0	1	0
8	11	5	2	0	0	0	0	1	0	0
8	17	9	1	0	0	0	0	1	0	0
8	18	9	2	1	1	0	0	0	1	0
9	3	2	1	0	0	0	0	1	0	0
9	12	6	1	0	0	0	0	1	0	0
9	33	12	1	1	1	1	0	0	1	0

12.5 ECG Effects

12.6 BP Effects

12.7 CVP Effects

12.8 Accel Effects

12.9 Most Common Combinations

12.10 Non-Typical Presentations

- PID 9, ARTID 3
- PID 9, ARTID 12

12.10.1 Misaligned Before Start

- PID 8, ARTID 10 - 2 mins before start shows a couple of BP artifacts
- PID 9, ARTID 33 - ECG, BP and CVP effects of Pupil testing?

12.10.2 Misaligned After Start

12.11 Algorithm In Words

12.12 Clinical Questions

- How often are pupils checked?

13 Compression_sock_adjust (N = 6, Number of Patients = 1)

13.1 Summary

Table 34: Artifact: Compression sock adjust Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Compression_sock_adjust	8	20	10	2	24.1 secs
	8	22	10	4	16.1 secs
	8	35	10	17	1 min 3.2 secs
	8	36	10	18	49.2 secs
	8	37	10	19	48.2 secs
	8	38	10	20	1 min 0.2 secs

13.2 Notes

Table 35: Artifact: Compression_sock_adjust Notes }

PID	ArtID	Task	Step	EPN	Note Text
8	20	10	2	-	CVP spikes showing a pumped infusion pattern.
8	22	10	4	-	No artifact within the annotation period. Adding +/- 0.5 mins shows some possible artifact on the ECG trace. There is a pumped infusion pattern in the CVP signal.
8	35	10	17	-	CVP spikes indicating that a pumped infusion is taking place. No noticeable artifact.
8	36	10	18	-	As artifact 35, no noticable artifact. CVP spikes indicating that a pumped infusion is taking place.
8	37	10	19	-	Similar to artifacts 35 and 36. No noticeable artifact. CVP spikes indicating that a pumped infusion is taking place.
8	38	10	20	-	Again, similar to the previous three ”Compression_sock_adjust” plots. No noticeable artifact. CVP spikes indicating that a pumped infusion is taking place.

13.3 Signal Summary

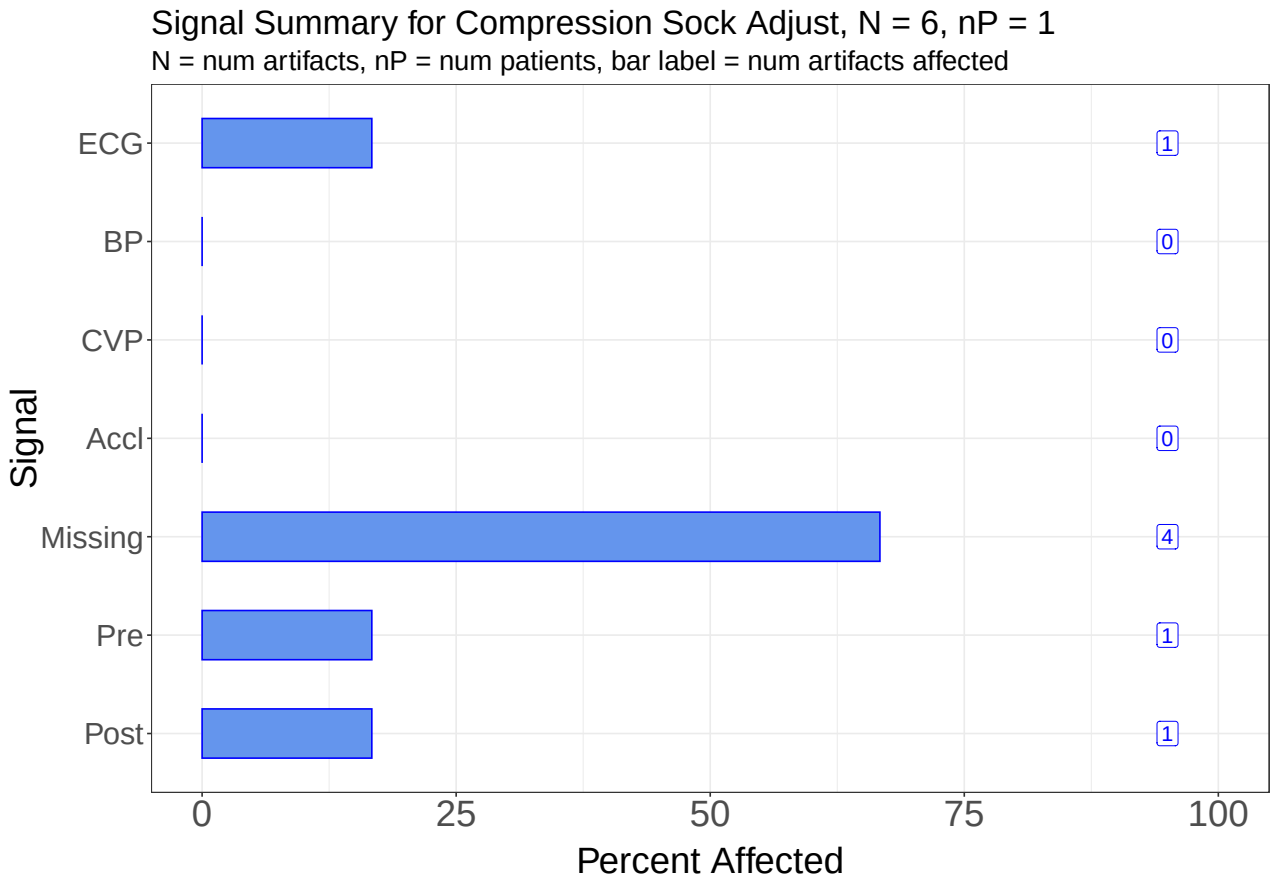

Figure 13: Artifact: Compression Sock Adjust Notes: Signal Summary

13.4 Notes Processing Summary

Table 36: Artifact: Compression Sock Adjust Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
8	20	10	2	0	0	0	0	0	0	0
8	22	10	4	1	0	0	0	0	1	1
8	35	10	17	0	0	0	0	1	0	0
8	36	10	18	0	0	0	0	1	0	0
8	37	10	19	0	0	0	0	1	0	0
8	38	10	20	0	0	0	0	1	0	0

13.5 ECG Effects

13.6 BP Effects

13.7 CVP Effects

13.8 Accel Effects

13.9 Most Common Combinations

13.10 Non-Typical Presentations

13.10.1 Misaligned Before Start

- PID 8, ARTID 22 +/- 0.5 mins see some ECG artifact

13.10.2 Misaligned After Start

- PID 8, ARTID 22 +/- 0.5 mins see some ECG artifact

13.11 Algorithm In Words

13.12 Clinical Questions

- As usual, how often does this occur?

14 Ng feed (N = 4, Number of Patients = 2)

14.1 Summary

Table 37: Artifact: Ng feed Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Ng_feed	8	7	2	1	1 min 6.3 secs
	8	8	3	1	19.1 secs
	8	16	8	1	1 min 21.3 secs
	9	35	14	1	19.1 secs

14.2 Notes

Table 38: Artifact: Ng feed Notes }

PID	ArtID	Task	Step	EPN	Note Text
8	7	2	1	-	This is an example of pumped infusion induced artifact (PIIA). You can see the pulses on CVP matching small pulses on ECG trace.
8	8	3	1	-	This is an example of pumped infusion induced artifact (PIIA). You can see the pulses on the CVP trace matching small pulses on ECG trace.
8	16	8	1	-	This is an example of pumped infusion induced artifact (PIIA). You can see the pulses on the CVP trace matching small pulses on ECG trace.
9	35	14	1	-	No obvious artifact on any of the signals.

14.3 Signal Summary

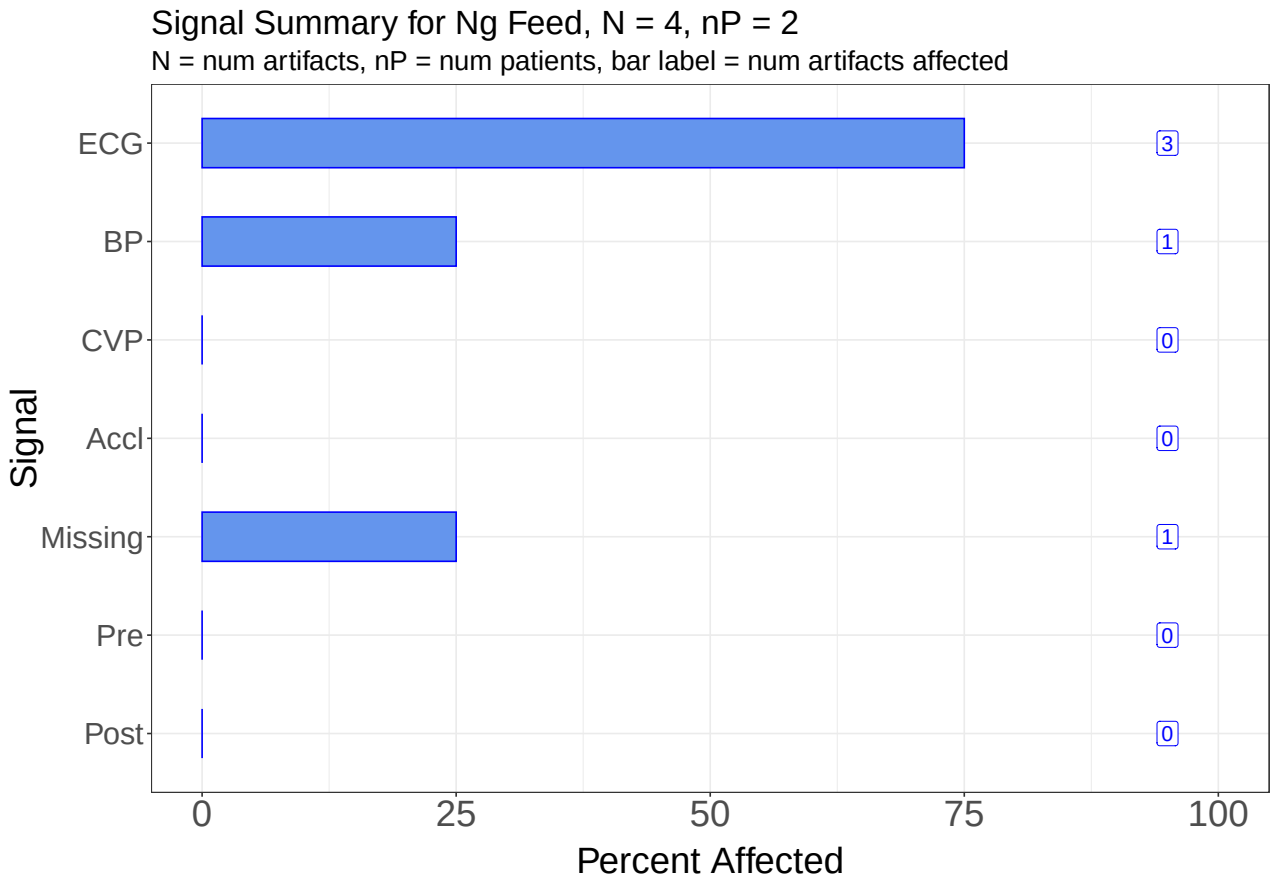

Figure 14: Artifact: Ng Feed Notes: Signal Summary

14.4 Notes Processing Summary

Table 39: Artifact: Ng Feed Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
8	7	2	1	1	1	0	0	0	0	0
8	8	3	1	1	0	0	0	0	0	0
8	16	8	1	1	0	0	0	0	0	0
9	35	14	1	0	0	0	0	1	0	0

14.5 ECG Effects

14.6 BP Effects

14.7 CVP Effects

14.8 Accel Effects

14.9 Most Common Combinations

14.10 Non-Typical Presentations

- PID 9 ArtID 35 - no obvious artifact… missed?

14.10.1 Misaligned Before Start

14.10.2 Misaligned After Start

14.11 Algorithm In Words

14.12 Clinical Questions

- How often is this occurring?

15 Temperature (N = 3, Number of Patients = 3)

15.1 Summary

Table 40: Artifact: Temperature Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Temperature	8	9	4	1	16.1 secs
	9	1	1	1	47.2 secs
	10	5	3	1	9 secs

15.2 Notes

Table 41: Artifact: Temperature Notes }

PID	ArtID	Task	Step	EPN	Note Text
8	9	4	1	-	No obvious artifact on any of the signals. Checked +/- 0.5 mins. Possible dips in CVP pattern.
9	1	1	1	-	No noticeable physiologic artifact. Definite accelerometer movement towards the end of the annotated period.
10	5	3	1	-	The annotation period shows a definite disturbance on the BP and CVP signals and maybe a dip in the ECG signal. There are no disturbances on the accelerometer signals during the annotation period. Looking at the 30 secs before the annotation start, there are definite disturbances on the ECG, BP, CVP and accelerometer Y signals.

15.3 Signal Summary

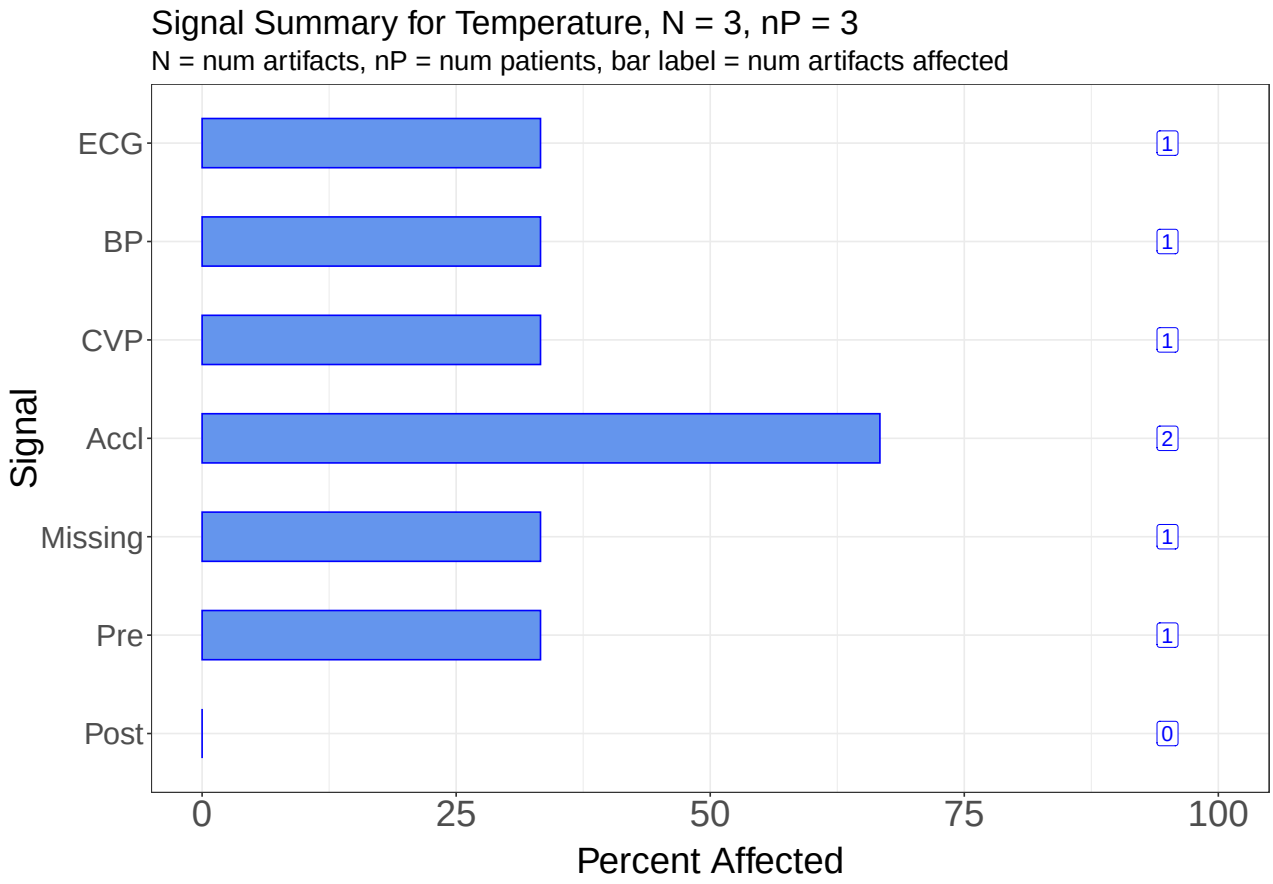

Figure 15: Artifact: Temperature Notes: Signal Summary **NOTE:** *Very little data*

15.4 Notes Processing Summary

Table 42: Artifact: Temperature Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
8	9	4	1	0	0	0	0	1	0	0
9	1	1	1	0	0	0	1	0	0	0
10	5	3	1	1	1	1	1	0	1	0

15.5 ECG Effects

15.6 BP Effects

15.7 CVP Effects

15.8 Accel Effects

15.9 Most Common Combinations

15.10 Non-Typical Presentations

- PID 8, ArtID 9 - no obvious effects
- PID 9, ArtID 1 - no obvious physiological effects. Accelerometer effects present.

15.10.1 Misaligned Before Start

- PID 10 ArtID 5 - 30 secs before effects on all channels

15.10.2 Misaligned After Start

15.11 Algorithm In Words

- inconsistent - not enough examples

15.12 Clinical Questions

- taking a patients temp with an ear probe. Maybe some jostling of the patient, might not see much?

16 Subglottic drainage (N = 3, Number of Patients = 2)

16.1 Summary

Table 43: Artifact: Subglottic drainage Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Subglottic_drainage	5	3	3	1	59.2 secs
	5	5	5	1	22.1 secs
	6	2	1	2	12.1 secs

16.2 Notes

Table 44: Artifact: Subglottic_drainage Notes }

PID	ArtID	Task	Step	EPN	Note Text
5	3	3	1	-	1. No obvious effects on any signals, slight drift in BP, but that is probably unrelated.
5	5	5	1	-	1. Clear changes in ECG/BP and x,y,z planes of accelerometer. No reason why this example of Subglottic_drainage is so different to the previous Subglottic_drainage event. Need to discuss with a clinician about the process of Subglottic_drainage and why they may be different.
6	2	1	2	-	1. Apart from the CVP signal ramping down from an infusion. No obvious changes in any signals. 2. Subglottic_drainage does seem to be quite variable in it's effects.

16.3 Signal Summary

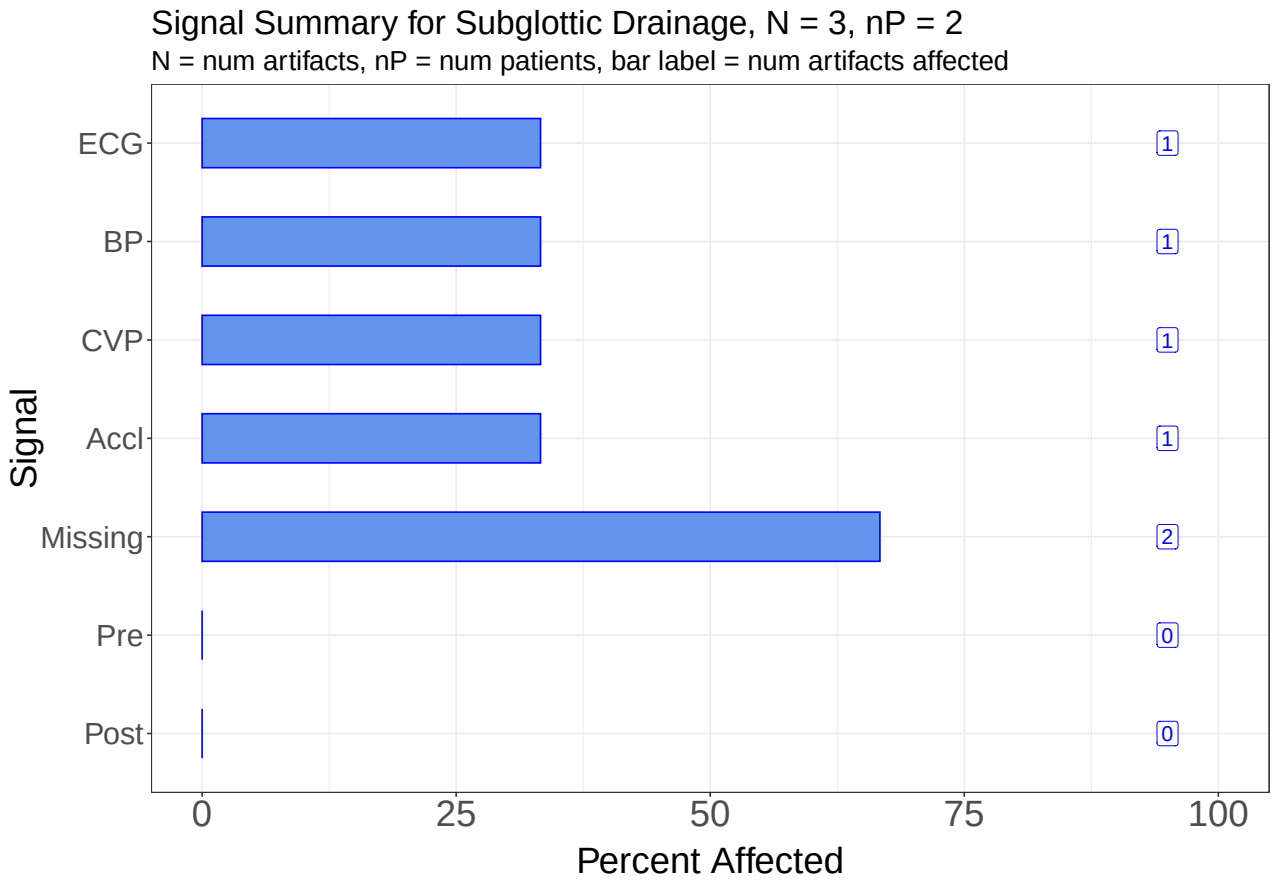

Figure 16: Artifact: Subglottic Drainage Notes: Signal Summary **NOTE:** *Very little data*

16.4 Notes Processing Summary

Table 45: Artifact: Subglottic Drainage Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
5	3	3	1	0	0	0	0	1	0	0
5	5	5	1	1	1	1	1	0	0	0
6	2	1	2	0	0	0	0	1	0	0

16.5 ECG Effects

16.6 BP Effects

16.7 CVP Effects

16.8 Accel Effects

16.9 Most Common Combinations

16.10 Non-Typical Presentations

- PID 5 ARTID 3 - no obvious effects
- PID 5 ARTID 5 - changes in all signals… why so different from others?
- PID 6 ARTID 2 - no obvious effects

16.10.1 Misaligned Before Start

16.10.2 Misaligned After Start

16.11 Algorithm In Words

- inconsistent effects half and half, not enough examples

16.12 Clinical Questions

- what clinically would you expect to see from sub-glottal drainage?

17 Patient moving (N = 2, Number of Patients = 2)

17.1 Summary

Table 46: Artifact: Patient moving Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Patient_moving	9	31	10	1	20.1 secs
	10	2	1	2	9 secs

17.2 Notes

Table 47: Artifact: Patient_moving Notes }

PID	ArtID	Task	Step	EPN	Note Text
9	31	10	1	-	No obvious artifact on any of the signals.
10	2	1	2	-	This is the end of the two step task. All signals apart from CVP seem to be stable during the annotation period. The CVP signal looks like all artifact. In particular it is at 188 to 189 mmHg which suggests that the transducer zeroing has been lost. During the minute before the annotation period there is disturbance on all signals apart from the artifactual CVP signal. Just after the annotation period there is movement on the accelerometer X and Y signals.

17.3 Signal Summary

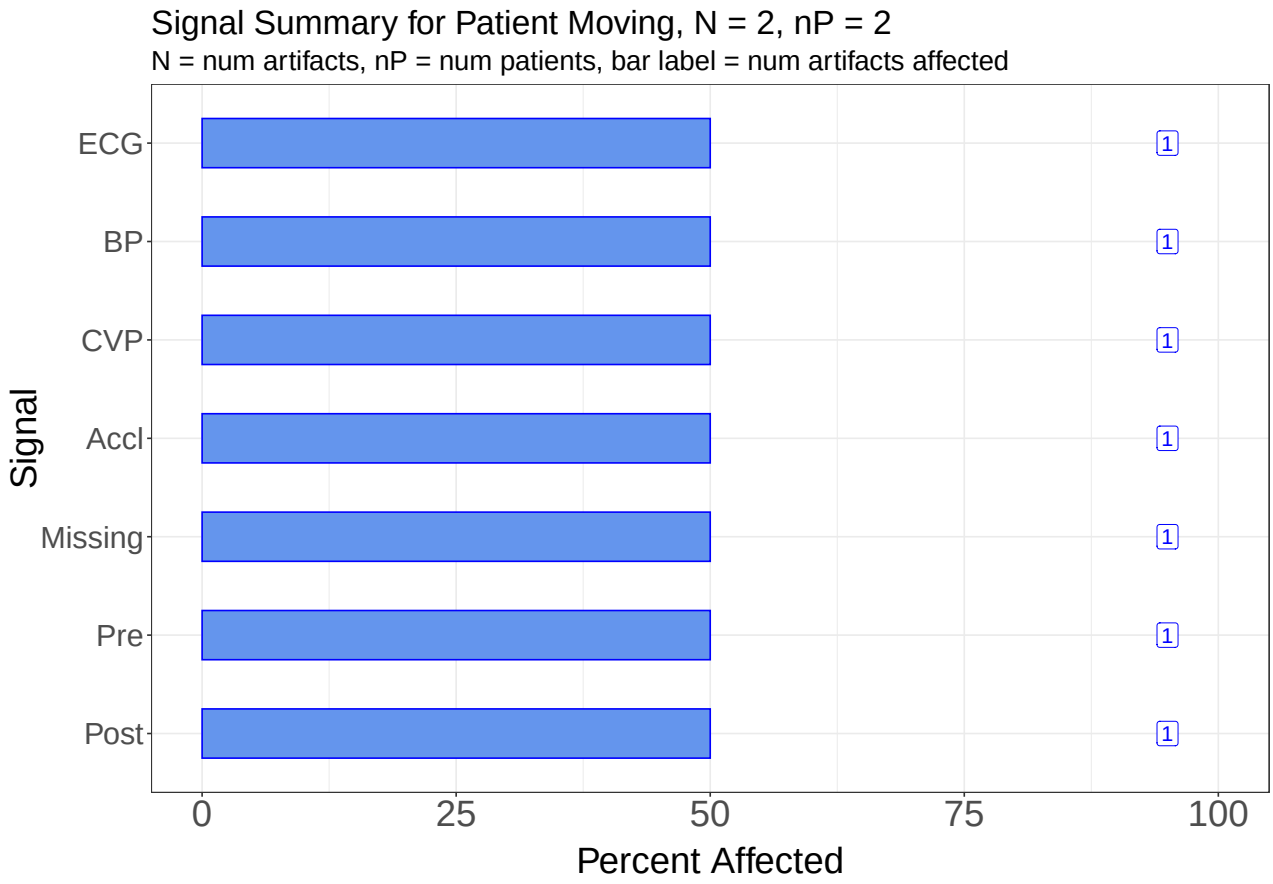

Figure 17: Artifact: Patient Moving Notes: Signal Summary **NOTE:** *Very little data*

17.4 Notes Processing Summary

Table 48: Artifact: Patient Moving Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
9	31	10	1	0	0	0	0	1	0	0
10	2	1	2	1	1	1	1	0	1	1

17.5 ECG Effects

17.6 BP Effects

17.7 CVP Effects

17.8 Accel Effects

17.9 Most Common Combinations

17.10 Non-Typical Presentations

- PID 9, ARTID 31 - no effects - missed?

17.10.1 Misaligned Before Start

- PID 10, ARTID 2, minute before noise on all signals

17.10.2 Misaligned After Start

- PID 10, ARTID 2, minute after noise on accel signals

17.11 Algorithm In Words

- Seems to affect all signals but timing may be off.

17.12 Clinical Questions

- is this a patient movement by nurses? or is the patient waking up?

18 Reposition limb (N = 2, Number of Patients = 2)

18.1 Summary

Table 49: Artifact: Reposition limb Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Reposition_limb	8	3	1	3	1 sec
	9	26	9	4	12.1 secs

18.2 Notes

Table 50: Artifact: Reposition_limb Notes }

PID	ArtID	Task	Step	EPN	Note Text
8	3	1	3	-	Very short (201*5) = 1.005 seconds
9	26	9	4	-	Artifact between the green and red cursors on ECG and BP signals. Possible artifact on BP signal just after the blue cursor. No obvious artifact on the accelerometer signals.

18.3 Signal Summary

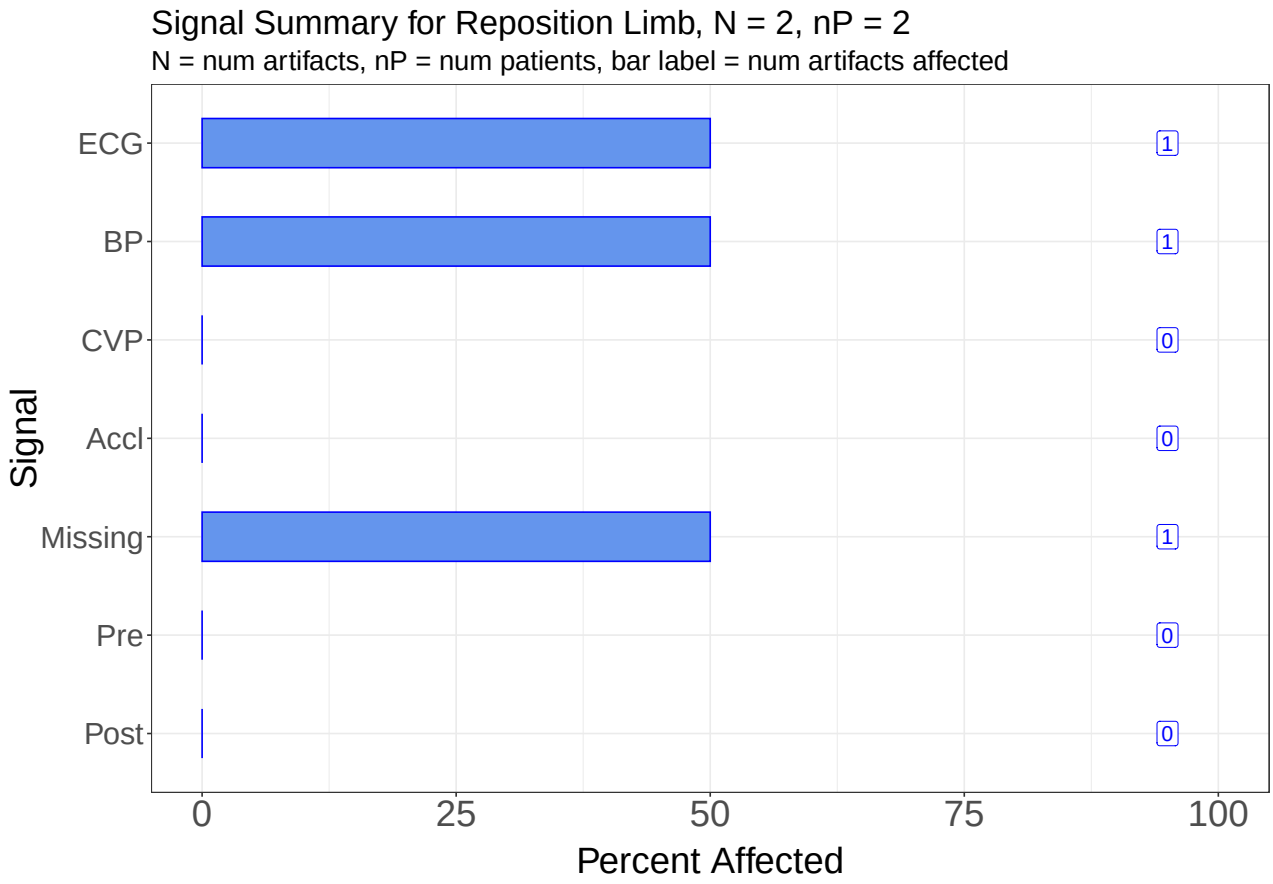

Figure 18: Artifact: Reposition Limb Notes: Signal Summary **NOTE:** *Very little data*

18.4 Notes Processing Summary

Table 51: Artifact: Reposition Limb Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
8	3	1	3	0	0	0	0	1	0	0
9	26	9	4	1	1	0	0	0	0	0

18.5 ECG Effects

18.6 BP Effects

18.7 CVP Effects

18.8 Accel Effects

18.9 Most Common Combinations

- very little effect, possibly an effect on BP

18.10 Non-Typical Presentations

- PID 8, ARTID 3 - very short event.

18.10.1 Misaligned Before Start

18.10.2 Misaligned After Start

18.11 Algorithm In Words

- Very little effect, possibly an effect on BP

18.12 Clinical Questions

- Would have expected to see some noise on Accel signals. Can one move limbs without jostling the body?

19 BP cuff (N = 2, Number of Patients = 1)

19.1 Summary

Table 52: Artifact: BP cuff Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
BP_cuff	9	6	3	2	29.1 secs
	9	8	3	4	8 secs

19.2 Notes

Table 53: Artifact: BP_cuff Notes }

PID	ArtID	Task	Step	EPN	Note Text
9	6	3	2	-	Artifact present on BP signal before the annotation start time. There appears to be still some artifact present at the start of the annotation period which settles maybe 25% through the period.
9	8	3	4	-	No artifact present on BP signal. Some possible artifact on the CVP signal which appears to start before the annotation start time.

19.3 Signal Summary

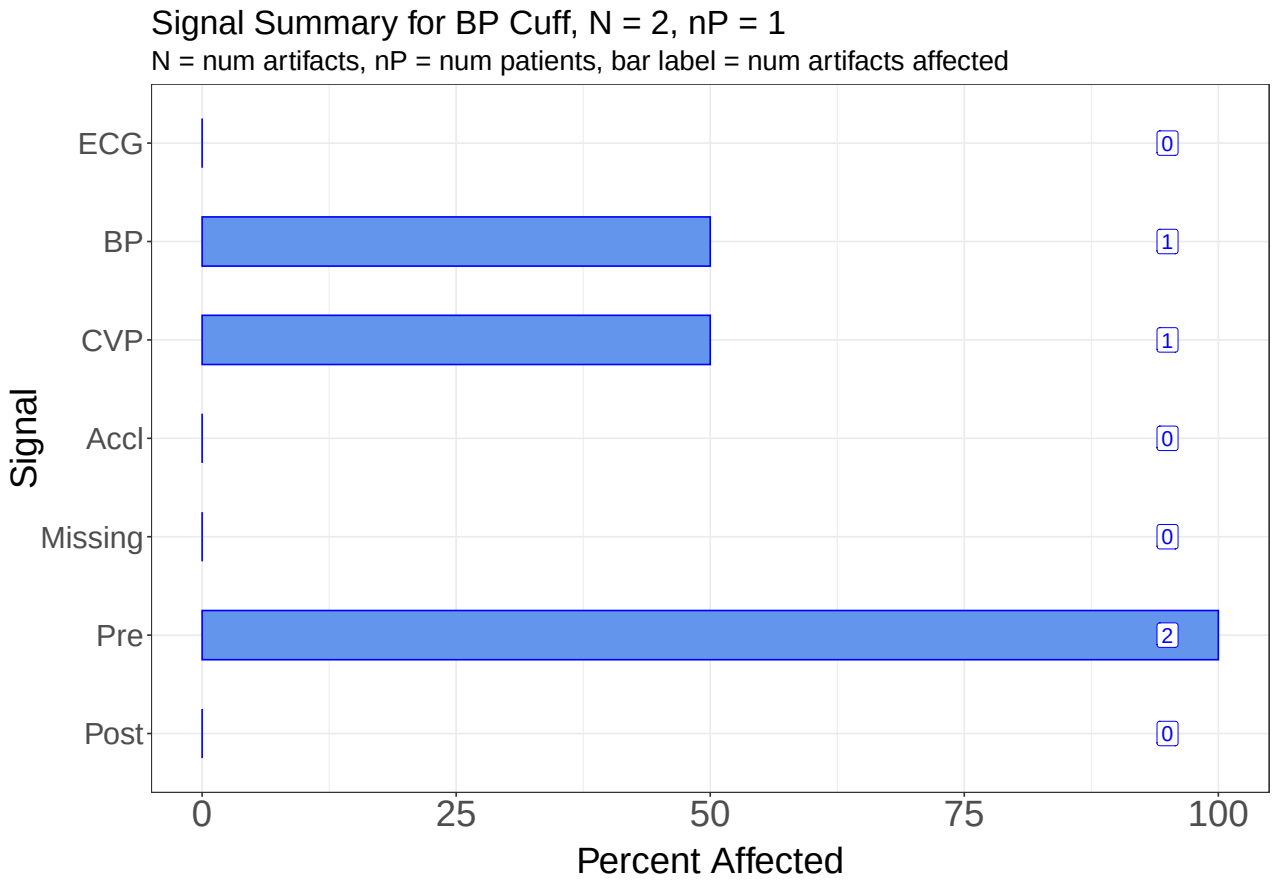

Figure 19: Artifact: BP Cuff Notes: Signal Summary **NOTE:** *Very little data*

19.4 Notes Processing Summary

Table 54: Artifact: BP Cuff Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
9	6	3	2	0	1	0	0	0	1	0
9	8	3	4	0	0	1	0	0	1	0

19.5 ECG Effects

19.6 BP Effects

19.7 CVP Effects

19.8 Accel Effects

19.9 Most Common Combinations

- some BP effects (to be expected), occasional CVP (handling arm?)

19.10 Non-Typical Presentations

19.10.1 Misaligned Before Start

- PID 9, ArtID 6 - some BP effects before
- PID 9, ArtID 8 - some BP effects before

19.10.2 Misaligned After Start

19.11 Algorithm In Words

- some BP effects (to be expected), occasional CVP (handling arm?)

19.12 Clinical Questions

- how often is the BP cuff BP checked? or is it only after invasive BP line removed?

20 Transducer calibration alteration (N = 2, Number of Patients = 1)

20.1 Summary

Table 55: Artifact: Transducer calibration alteration Summary

Artifact Type	Patient ID	Artifact Number	Task	Step	Step Time
Transducer_calibration_alteration	9	7	3	3	17.1 secs
	9	24	9	2	14.1 secs

20.2 Notes

Table 56: Artifact: Transducer_calibration_alteration Notes }

PID	ArtID	Task	Step	EPN	Note Text
9	7	3	3	-	Distinct artifact on BP and CVP signals. No apparent movement on the accelerometer signals.
9	24	9	2	-	Noticeable BP signal disturbance, pre, during and after the annotation period. The CVP signal shows an artifact pattern in the pre-annotation period at the green cursor. It has to be said though that the green cursor is only 8 seconds before the annotation start time.

20.3 Signal Summary

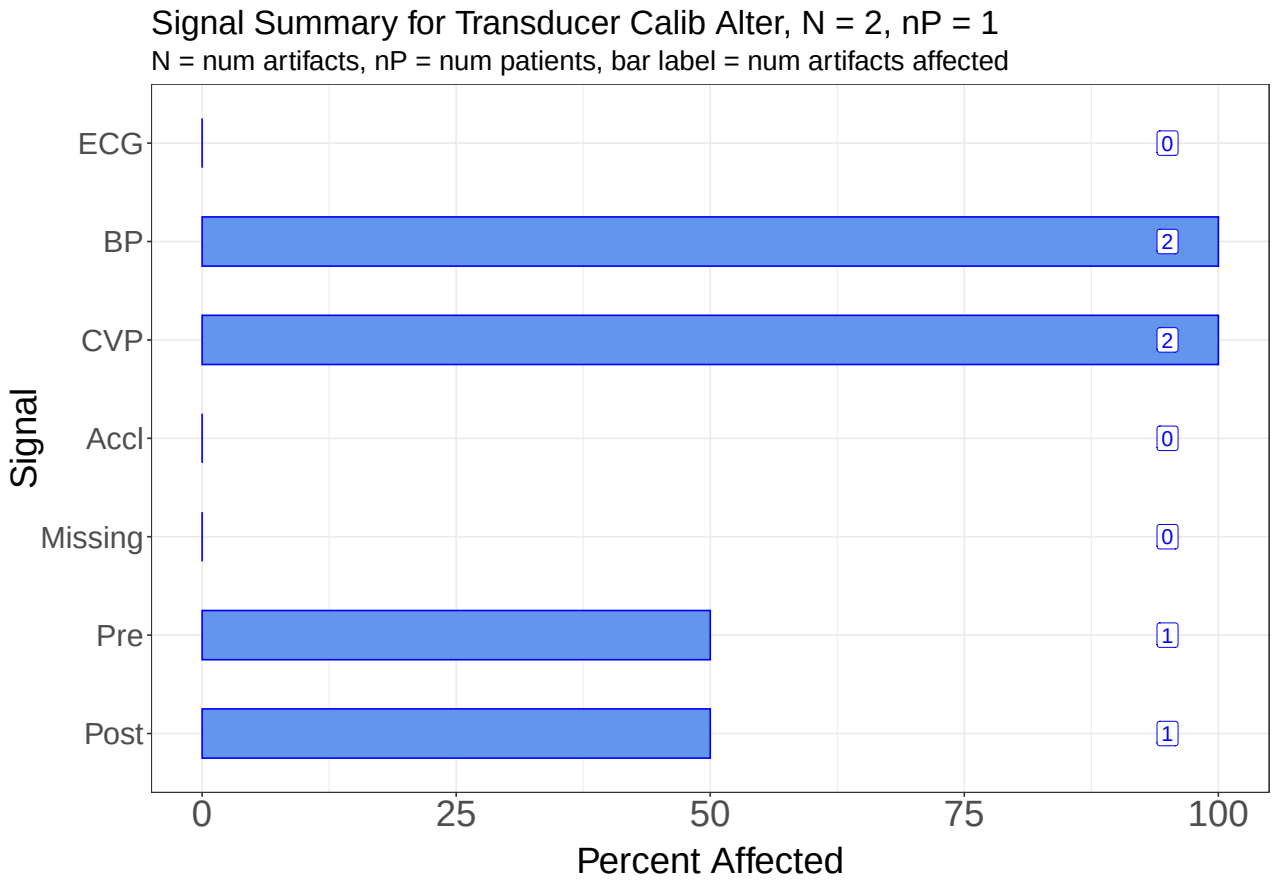

Figure 20: Artifact: Transducer Calibration Alteration Notes: Signal Summary **NOTE:** *Very little data*

20.4 Notes Processing Summary

Table 57: Artifact: Transducer Calibration Alteration Notes Processing Summary

PID	ArtID	Task	Step	ECG	BP	CVP	Accl	Miss	Pre	Post
9	7	3	3	0	1	1	0	0	0	0
9	24	9	2	0	1	1	0	0	1	1

20.5 ECG Effects

20.6 BP Effects

20.7 CVP Effects

20.8 Accel Effects

20.9 Most Common Combinations

- BP and CVP noise

20.10 Non-Typical Presentations

20.10.1 Misaligned Before Start

- PID 9 ARTID 24 - 8 second before start CVP noise.

20.10.2 Misaligned After Start

20.11 Algorithm In Words

- BP and CVP noise

20.12 Clinical Questions

- how often? Once per nursing shift? Just BP transducer or is CVP transducer zeroed too?

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